



# **INTERNSHIP REPORT**

**ON**

## **9W LED BULB MANUFACTURING PROCESS**

**Submitted By:**

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**Internship Duration:**

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**Submitted To:**

Colorhome Smart Lighting Pvt. Ltd.

**Date of Submission:**

26<sup>th</sup> August, 2025

**In supervision of:**

Mr. Pramesh Gupta

Mr. Narsingh Prajapati

Mr. Sachin Tyagi

## **ACKNOWLEDGEMENT**

I would like to express my heartfelt gratitude to Mr. Pramesh Gupta for granting me this valuable opportunity to undertake my internship. His unwavering support, trust, and timely guidance played a pivotal role in shaping this experience. He not only ensured that I was paired with the right mentor but also consistently followed up to make sure the internship remained productive, insightful, and aligned with my learning objectives.

I am deeply thankful to Mr. Narsingh Prajapati for being a constant source of technical knowledge and encouragement. His hands-on guidance throughout the internship helped me understand the manufacturing process of the 9W LED bulb in depth. His willingness to clarify concepts and share his expertise greatly enhanced my understanding and technical confidence.

I would also like to thank Mr. Sachin Tyagi for accompanying me during the course of my internship and assisting me in preparing this report. His supervision and support helped me resolve doubts and avoid errors throughout the documentation process.

I extend my gratitude to the entire team at Colorhome Smart Lighting Pvt. Ltd., particularly in the Design and Development Department, for their cooperation and support during my internship.

Finally, I thank my faculty and PES University, RR Campus, Bengaluru, for facilitating this internship and for their continuous academic support.

This experience has greatly contributed to my practical knowledge and professional growth.

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## **COLORHOME SMART LIGHTING – COMPANY PROFILE**

### **About the Company**

Founded in 2020 under Sunpu India LED Co. LLP and headquartered in Noida, Colorhome Smart Lighting develops innovative, sustainable lighting solutions tailored for homes and businesses across India. Embracing the philosophy that “business is an extended family,” the company prioritizes ethical operations, stakeholder-centric values, and eco-friendly engineering.

### **Vision & Mission**

- Vision: Reach every household in India with affordable, quality lighting products.
- Mission: Conduct ethical, transparent business while providing reliable, cost-effective lighting.

### **Core Products & Offerings**

- LED Bulbs & Lamps: Hybrid (3 W–30 W), 9 W color/RGB bulbs, emergency and DC-compatible bulbs
- LED Battens & Panels: Ceiling and surface-mounted luminaires for both residential and commercial use
- Outdoor & Industrial Lighting: Streetlights, floodlights, and high-bay fixtures
- Solar Products: Solar-powered street and flood lighting for off-grid areas
- Smart Lighting Solutions: IoT-enabled bulbs, strips, and luminaires for automated control
- Accessories & Emergency Lighting: Plugs, holders, spike guards, torches, lanterns, and battery-backed LEDs

### **Value Propositions**

- Energy Efficiency: Cutting-edge LED design reduces power usage
- Durability & Safety: LM80-tested LEDs, IEC-compliant optics, and heat-sink drivers ensure longevity and safety
- Design Variety: A diverse product portfolio from decorative RGB bulbs to robust industrial fixtures
- Customer-Centric Policies: Ethical business practices including free shipping on orders over ₹1000 and hassle-free returns

### **Internship Placement**

I completed my internship with the Design and Development Department at Colorhome Smart Lighting Pvt. Ltd., where I worked on 9W LED bulb design, understanding component selection, thermal solutions, and end-to-end manufacturing flows. This hands-on experience deepened my technical expertise and aligned with Colorhome’s mission of sustainable innovation.

## **PROJECT INTRODUCTION**

During my internship at Colorhome Smart Lighting Pvt. Ltd., I had the opportunity to work on a project that focused on understanding the complete design and manufacturing process of a 9W LED bulb. I was placed in the Design and Development Department, where I gained hands-on experience with one of the most widely used and efficient lighting products in the market today.

The goal of the project was to observe and document how a 9-watt LED bulb is built—from start to finish. At first glance, a bulb may seem like a simple household item, but through this internship, I learned how much engineering, precision, and quality control goes into producing a single unit. The 9W bulb in particular is an essential product because it offers a balance of brightness and energy efficiency, making it a preferred choice for homes, offices, and public spaces.

My responsibilities included learning about and analyzing the components that make up the bulb—such as the LED chips, PCB board, driver circuit, metal oxide varistor (MOV), fuse, heat sink, diffuser, and the plastic and metal housing. I observed each step of the production line, including component assembly, soldering, testing, thermal paste application, and quality inspection. I also got to see how ageing tests are conducted and how the branding is applied using laser printing machines.

Working alongside engineers and production staff helped me understand not just the technical aspects, but also the importance of teamwork, attention to detail, and maintaining high safety and quality standards. I also learned how every small component plays a vital role in the performance and safety of the final product.

What made the project even more interesting was seeing how the company integrates smart features and modern design into traditional lighting. It gave me a broader view of where the lighting industry is headed—towards smarter, more sustainable, and user-friendly products.

This report is a reflection of everything I learned during my time at Colorhome. It covers both the technical side of the 9W bulb and the practical lessons I picked up by being part of a real production environment. It was a rewarding experience that helped me apply my classroom knowledge in a real-world setting, and I'm excited to share it here.

# 9W LED Bulb Hybrid A60 16B – Detailed Component Report

## Introduction

This report provides a clear and logical description of the key components that make up a 9W Driver-on-Board (DOB) LED bulb. Each part has been carefully selected to ensure performance, durability, safety, and efficiency. For ease of understanding, the components are grouped into electronic parts, which handle power and control, and mechanical/packaging parts, which ensure structure, protection, and appearance.

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## Electronic Components

### 1. Metal Core PCB (MCPCB – PMSBU24430 VER.A)

A 1.0 mm thick aluminum-based board with a diameter of 39.8 mm and a 35-micron copper layer. It is specifically designed for high-power LED applications, providing both a mechanically stable mounting surface and excellent heat dissipation. The copper layer ensures low-resistance electrical paths, while the aluminum base effectively transfers heat away from the LEDs, improving reliability and lifespan.

### 2. LED (1W, 17–19V, 60mA, 150–160lm, 6500K, Ra-80, SDCM<5)

These high-efficiency LEDs are the main light source of the bulb. Operating at low current, each produces a bright cool-white output of 150–160 lumens with a color temperature of 6500K and a color rendering index of 80, ensuring natural color reproduction. The SDCM<5 rating means color consistency across LEDs, avoiding any visible color differences when multiple LEDs are used together.

### 3. SMD Resistors (2M $\Omega$ ±5%, 0 $\Omega$ ±5%, 10 $\Omega$ ±1%)

- **2M $\Omega$ :** Used to bleed residual voltage or provide high-impedance paths, ensuring safety after power-off.

- **0 $\Omega$ :** Acts as a PCB jumper or link, simplifying layout while maintaining flexibility for circuit modifications.

- **10 $\Omega$ :** Utilized in series with circuits to limit current spikes, improve stability, or suppress noise, protecting sensitive ICs. These tiny surface-mounted components are crucial for precision control and reliability in the compact PCB design.

### 4. Bridge Rectifier (MB10F, 1000V/1.0A)

This miniature full-wave bridge rectifier converts AC input into DC for the driver circuit. It is rated for 1000V reverse voltage and 1A forward current, ensuring safe operation under standard mains conditions. Its low-profile SMD package allows compact design while providing reliable rectification.

#### **5. Connector (2P-F007 Female)**

A two-pin female connector designed to provide a secure interface between the driver PCB and external wiring. It is built for stable electrical contact and to withstand vibrations or handling during installation and maintenance.

#### **6. Driver IC (BP5336HC, E-SOP-8)**

This integrated circuit is the heart of the driver section. It maintains a constant current to the LEDs, ensuring steady light output. The IC features over-voltage protection, thermal shutdown, and short-circuit protection, improving both safety and longevity. Its compact E-SOP-8 package enables efficient use of space on the MCPCB.

#### **7. Solder Paste (Sn/Pb 63/37)**

A tin-lead eutectic solder paste with a composition of 63% tin and 37% lead, known for its excellent wetting properties and reliable bonding. When reflowed, it creates robust solder joints that can withstand years of thermal cycling and vibration without failure.

#### **8. Fuse Resistor (100Ω ±5%, 1W, 65 Tube 46mm, L=4+15mm)**

This component serves two purposes: it limits inrush current during startup and acts as a safety fuse in case of overload or short circuit. Its physical size and specification (100Ω, 1W) allow it to handle normal current flow while reliably breaking the circuit under fault conditions.

#### **9. Metal Oxide Varistor (7D621K-A, 105°C, P=5mm, L=5mm)**

A surge-protection device that safeguards the circuit from high-voltage spikes. With a lead spacing of 5mm and a compact body, it reacts quickly to transient over-voltages by clamping the voltage and dissipating excess energy as heat, protecting the sensitive electronics.

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### **Mechanical & Packaging Components**

#### **10. A60 16B Housing (240°)**

A precisely molded polycarbonate shell that forms the outer shape of the bulb. It supports internal parts, manages heat, and ensures that light is distributed evenly over a 240-degree beam angle.

#### **11. A60 16B Diffuser (240°)**

A translucent cover that softens the LED light, reducing glare and providing pleasant, uniform illumination throughout the room.

#### **12. B22 Cap (Aluminum Rivet Type)**

The standard bayonet cap that allows the bulb to fit into common sockets. Made of corrosion-resistant aluminum, it ensures a secure electrical connection.

### **13. Mono Insert (Protective Piece)**

A custom protective insert placed around each bulb during packaging to prevent movement and reduce damage during transport.

### **14. Master Carton**

The outer shipping box that holds multiple bulbs. It is designed for maximum protection and space efficiency during storage and shipping.

### **15. Weight Sticker**

A simple label on the packaging that displays the product's weight. This assists with logistics, handling, and inventory management.

### **16. BOPP Tape**

A strong adhesive tape used to seal individual boxes and master cartons, ensuring they remain secure during transit.

### **17. Stripping Roll (0.9 mm)**

A band used to bundle and secure multiple boxes or cartons together on pallets, improving safety and efficiency in bulk transport.

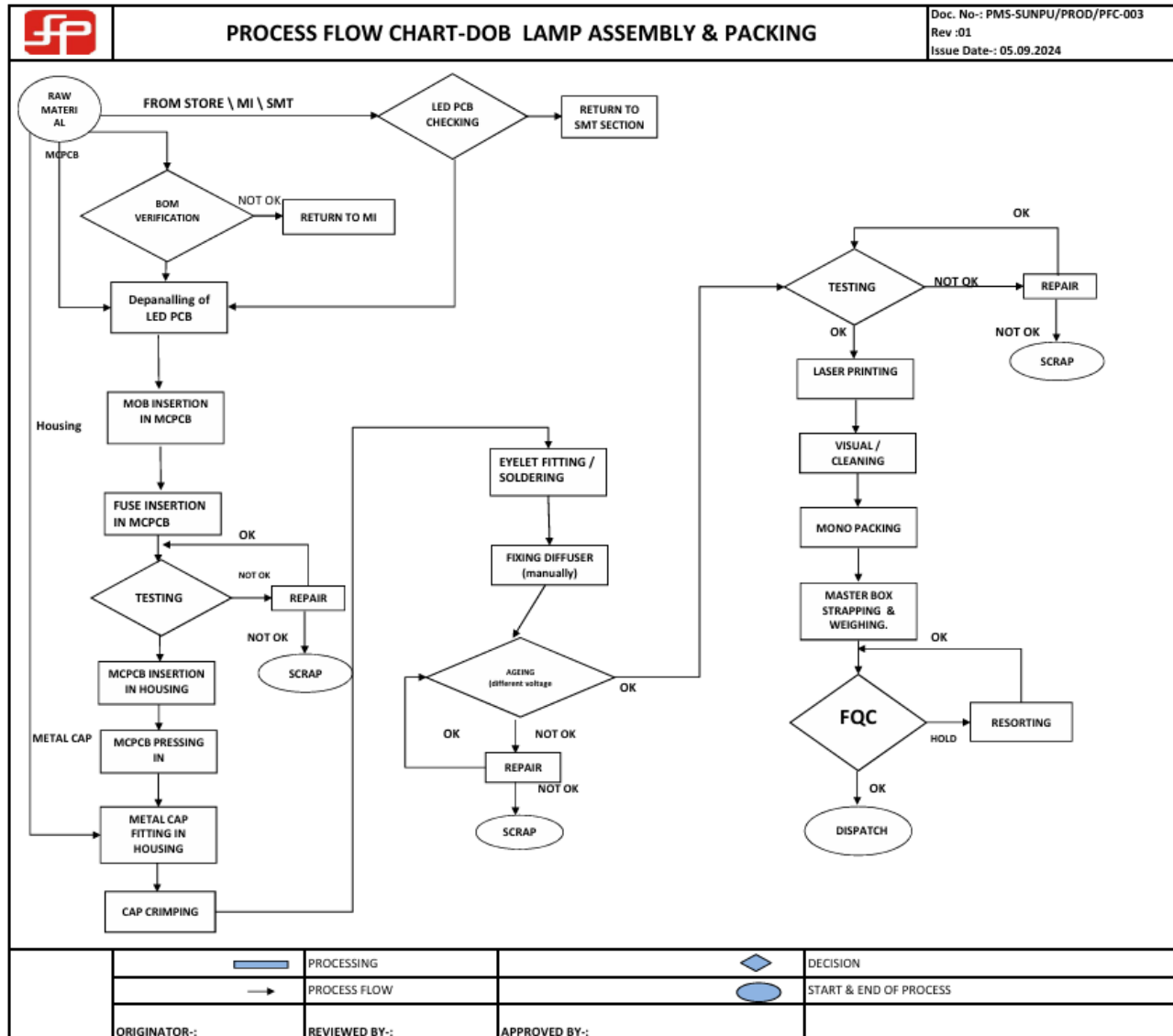
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## **Conclusion**

Each electronic component described above plays a crucial role in ensuring the 9W LED DOB bulb operates efficiently, safely, and reliably. From the thermally optimized MCPCB and carefully binned LEDs to the surge-protecting MOV and current-regulating driver IC, these elements work together seamlessly. Combined with thoughtful mechanical design and packaging, the result is a durable product that delivers consistent performance and user satisfaction over its lifetime.

# MANUFACTURING OF A DOB 9-WATT BULB

Following is a flow chart for the manufacturing of the DOB 9-watt bulb:





The process of manufacturing in Colorhome is divided into 3 sections:

1. SMT – system mount technology  
The components which have to be mounted on MCPCB which cannot be done by hand or would take a lot of time by hand are automated and have no human intervention.
2. MI – Manual Insertion  
The manual insertion refers to the work which has to be done by hand or human operated machineries.
3. Assembly  
There are assembly lines in the factory where most of the non-technical parts are assembled together.

As DOB (driver on board) 9-watt bulb does have much in the manual insertion, it is also done on the assembly.

## BOM verification:

The process starts with bill of material verification, verifying if all the components which have been imported or have been acquired from raw material providers have been sent in one piece or not.

## System Mount Technology:

Following are the devices that go through the SMT process:

| Sl.NO | Item Descriptions                                                                                      | Package | Location | QTY           |
|-------|--------------------------------------------------------------------------------------------------------|---------|----------|---------------|
| 1     | Metal core PCB , Thickness -1.0 mm ,<br>Dia-39.8 mm, 35micron Cu<br><a href="#">[PMSBU24430 VER.A]</a> |         |          | 1 pcs         |
| 2     | LED,1W, 17 to 19V, 60mA, 150-160lm,<br>6500K, Ra-80 SDCM <5                                            | 2835    | MCPCB    | 15 pcs        |
| 3     | SMD Resistors 2M $\Omega$ ±5%                                                                          | 1206    | R1       | 1 pcs         |
| 4     | SMD Resistors 0 $\Omega$ ±5%                                                                           | 1206    | R2       | 1 pcs         |
| 5     | SMD Resistors 10 $\Omega$ ±1%                                                                          | 805     | RS1      | 1 pcs         |
| 6     | MB10F, 1000V/1.0A                                                                                      | SOP-4   | BR1      | 1 pcs         |
| 7     | CONNECTOR_2P-F007_Female                                                                               | F007    | T1,T2    | 2 pcs         |
| 8     | IC-BP5336HC E-SOP-8                                                                                    | SOP-8   | U1       | 1 pcs         |
| 9     | Solder Paste , Sn/Pb 63/37                                                                             |         |          | 0.00012<br>kg |

Location refers to the location on the pcb which is designed by the pcb designers in the R&D department. Packages are referred to the specific packages they come in.

## **Surface Mount Technology (SMT) Process:**

The Surface Mount Technology (SMT) process is a core technique in modern electronics manufacturing and is especially important in the production of 9-watt LED bulbs. SMT enables precise placement and soldering of miniature electronic components directly onto the Metal Core Printed Circuit Board (MCPCB). Each stage of the process is designed to improve manufacturing quality, electrical reliability, and overall product durability. For engineering students and professionals, understanding SMT is essential because it integrates design, materials science, and automated assembly into one workflow.

### **1. Glue Dispenser Machine**

In the first stage, solder paste or adhesive is applied to the MCPCB at specific component locations. For a 9-watt LED bulb, solder paste is preferred because it not only secures the component temporarily but also establishes the final electrical connection during reflow. The glue dispenser machine, often semi-automated, must deposit the correct amount of paste with high positional accuracy. Uneven application can lead to poor joints or unwanted bridging, so operators carefully monitor the process to ensure consistency and repeatability.

#### **Process Instructions:**

- I. While working on the glue machine, follow these important points.
- II. Clean the machine's blade and stencil twice daily — once before starting the shift and again during changeover.
- III. Use thinner to clean the blade and stencil thoroughly.
- IV. Ensure the blade doesn't stick to the stencil.
- V. Change the blade after 45–60 days.
- VI. Replace the stencil regularly based on its condition and usage.
- VII. Air pressure in the glue machine should be maintained at 4–5 kg/cm<sup>2</sup>.

|                                                                                                                                   |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                    |  |                         |  |
|-----------------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------|--|-------------------------|--|
|  <b>कार्य निर्देश</b><br><b>WORK INSTRUCTION</b> |  | <b>MODEL- ALL MODEL</b>                                           |  | Process:- ग्लू मशीन और<br>स्टेसिल कि सफाई                                                                                                                                                                                                                                                                                                                                                                                                                           |  | Prepared by<br>Ujjwal Srivastav                                                    |  | Approved by<br>Ravi Jha |  |
|                                                                                                                                   |  |                                                                   |  | Doc. No.: PMS SUNP/UP/PC/OW/11                                                                                                                                                                                                                                                                                                                                                                                                                                      |  |                                                                                    |  |                         |  |
|                                                                                                                                   |  |                                                                   |  | Issue Date/Rev. No. 08/07/2023/00                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |                                                                                    |  |                         |  |
| <b>Material</b>                                                                                                                   |  | <b>Qty</b>                                                        |  | <b>प्रसंस</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  | <b>डाटाग्राम</b>                                                                   |  |                         |  |
| पेस्ट/ ग्लू<br>मशीन                                                                                                               |  | प्लान के<br>अनुसार<br>As per<br>plan                              |  | 1 ग्लू मशीन पे काम करते हुए निम्न बातों का ध्यान रखें. ग्लू<br>2 मशीन के blade और स्टेसिल को दिन में दो बार शिफ्ट शुरू होने<br>से पहले और changeover के टाइम अवश्य साफ करें.<br>3 Blade और स्टेसिल को थिनर कि सहायता से अच्छे से साफ करें<br>ग्लू कहीं blade पर ना लगा रह जाए<br>4 45-60 दिन बाद ब्लेड चेंज कर दें<br>5 स्टेसिल को भी टाइम टाइम पर उसकी कंडीशन के हिसाब से<br>6 change करते रहें.<br>ग्लू मशीन का Air pressure 4-5 kg/cm <sup>2</sup> . होना चाहिए। |  |  |  |                         |  |
| <b>Equipment Used</b>                                                                                                             |  | <b>सावधानी PRECAUTION</b>                                         |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                    |  |                         |  |
| मैनुअल                                                                                                                            |  | कार्य शुरू करने से पहले फिंगर टिप्स और इ.स.डी बैंड जरूर<br>पहनें. |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                    |  |                         |  |
| <b>सुरक्षा बिंदु SAFTY POINTS</b>                                                                                                 |  | अगर कुछ गलत हो रहा है<br>going any wrong                          |  | #                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  | Revision History                                                                   |  |                         |  |
| ESD BAND, SLIPPER                                                                                                                 |  | अपने लाइन प्रभारी को सूचित<br>करें                                |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  | No. Date Reason Approval No. Date Reason Approval                                  |  |                         |  |
|                                                                                                                                   |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                    |  |                         |  |
|                                                                                                                                   |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |                                                                                    |  |                         |  |

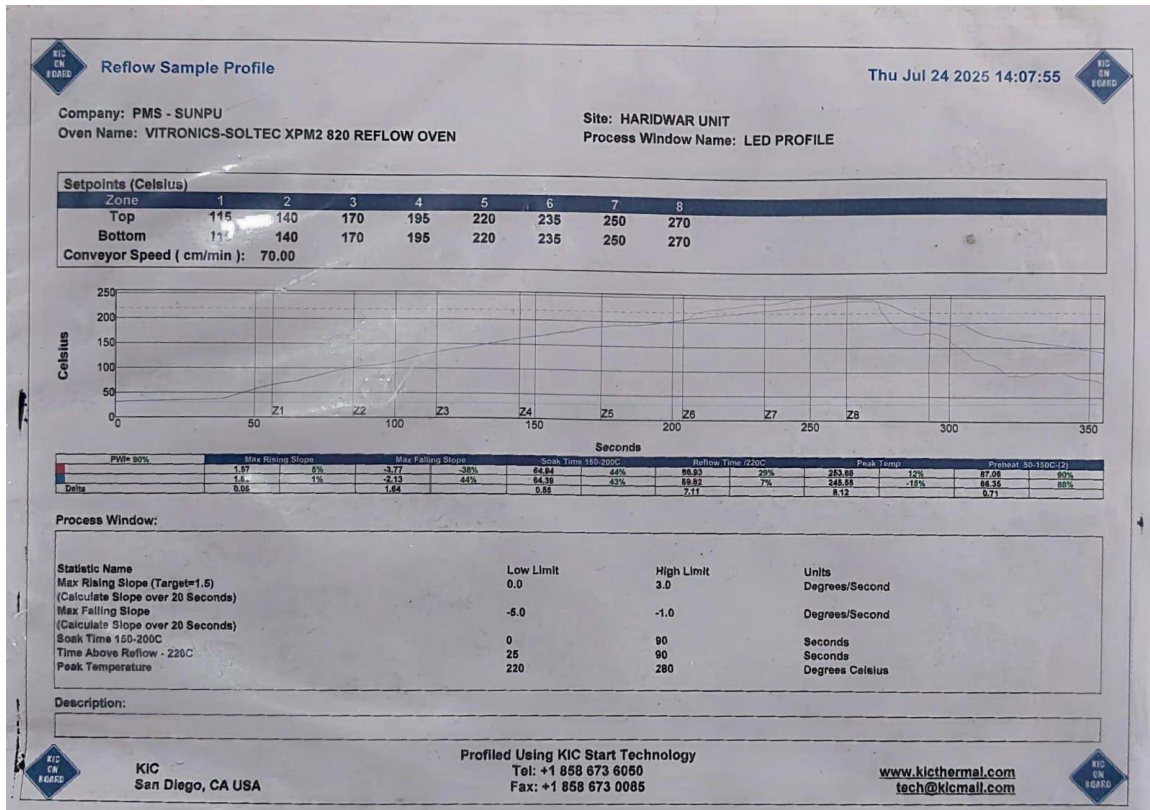
## 2. Mounting Machine

After solder paste application, the MCPCB is transferred to the mounting machine. This high-precision equipment uses pick-and-place mechanisms to accurately position components such as LEDs, resistors, and driver ICs onto the prepared pads. The machine operates based on the board's digital design files, enabling it to handle various layouts and component sizes with minimal human intervention. For a 9-watt bulb, correct placement is critical to achieve uniform light output and stable electrical performance. Regular calibration and preventive maintenance are necessary to maintain accuracy and throughput.

|  <b>WORK INSTRUCTION</b> |                                                                                                                                                                                                                                                                                                                                                                                          | Doc No:- PMS SUNPU/PROD/SMT/WI/002<br>Issue Date:-27.07.2017<br>Revision :- 00 |
|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>1. TITLE</b>                                                                                           | Work instruction for Operation and Preventive Maintenance Of Yamaha YV100X.                                                                                                                                                                                                                                                                                                              |                                                                                |
| <b>2. SCOPE</b>                                                                                           | This Work Instruction is applicable for Pick & place of SMDs to PCB.                                                                                                                                                                                                                                                                                                                     |                                                                                |
| <b>3. REFERENCE</b>                                                                                       | 3.1 : Yamaha YV100X Operating Manual<br>3.2 : Yamaha YV100X User Reference Manual<br>3.3 : Yamaha YV100X Service Manual                                                                                                                                                                                                                                                                  |                                                                                |
| <b>4. CONSUMABLES/TOOLS/ EQUIPMENTS</b>                                                                   | <div> <div> 4.1 : Consumables : Cleaning Paper, Solvent, Grease<br/> 4.2 : Tools : Allen Key, Screw Driver Set, Vernier Calliper, Feeder Loading Unit.<br/> 4.3 : Equipment : Yamaha YV100X Machine </div>   </div> |                                                                                |
| <b>5. APPLICABLE FORMAT</b>                                                                               | Feeder Loading Sheet                                                                                                                                                                                                                                                                                                                                                                     |                                                                                |
| <b>6. PROCEDURE</b>                                                                                       | 6.1 : Power ON machine<br>6.2 : Open Air Pressure, set it to 5.7 Bar<br>6.3 : Set the Machine to Origin<br>6.4 : To Run, Select the Program & Load Components according to Model                                                                                                                                                                                                         |                                                                                |
| <b>7. OPERATION</b>                                                                                       | <div> <div> 7.1 : Prepare the required Program on Machine<br/> 7.2 : Get all the Comp. recognised by machine using Comp. Database<br/> 7.3 : Optimize the program </div> <div> 7.4 : Load all the components according to loading sheet<br/> 7.5 : Make first PCB for Detail Analysis with BOM<br/> 7.6 : If there is shifting &amp; missing, Modify Program </div> </div>               |                                                                                |
| <b>8. PRECAUTIONS</b>                                                                                     | 8.1 : Check ground to neutral voltage on daily basis with the help of maintenance dept.<br>8.2 : Clean the machine as per daily check list before in use.<br>8.3 : Do not operate the machine with By-Pass Safty Functions, Follow Warning Stickers.                                                                                                                                     |                                                                                |
| <b>9. MAINTENANCE</b>                                                                                     | 9.1 : General cleaning of machine on daily basis.<br>9.2 : Preventive maintenance as per schedule.                                                                                                                                                                                                                                                                                       |                                                                                |
| PREPARED BY :                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                          | APPROVED BY :                                                                  |

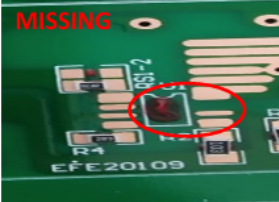
### 3. Reflow Soldering

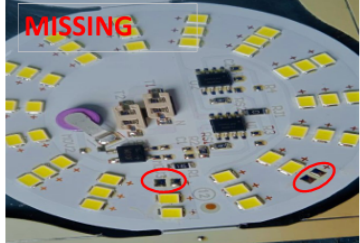
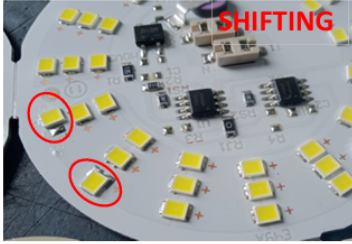
Once components are mounted, the board proceeds through the reflow oven. Here, it passes through multiple temperature zones that gradually raise the board temperature until the solder paste melts, creating strong metallic bonds between component leads and the copper pads of the MCPCB. The heating profile is carefully engineered to avoid damaging heat-sensitive parts while ensuring a complete reflow. After cooling, these joints provide both mechanical support and reliable electrical connections, which are vital for the long-term operation of a 9-watt LED bulb.



## Visual Inspection and Quality Control

The final stage involves meticulous visual inspection and quality control. Skilled workers examine each board for misaligned components, insufficient solder coverage, or parts displaced during reflow. Because automated inspection systems are not always used in smaller production setups, human inspection plays a critical role. If any defects are found, technicians rework the board or replace faulty components. In some facilities, additional functional testing is performed to verify that the circuit operates as intended. Only boards that meet strict quality standards proceed to final assembly, ensuring that every 9-watt LED bulb delivered to customers is safe, reliable, and built to last.

|                                                                                                                            |            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                       |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
|----------------------------------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------|--------------------------------|------|--------|----------|
|  <b>कार्य निर्देश</b><br>WORK INSTRUCTION |            | <b>Model :- All Model of Driver</b>                                                                                 |                                                                                                                                                                                                                                                                                                                       | <b>प्रोसेस :-</b><br>SMT VISUAL INSPECTION |                                                                                                                                                                                                                                                             | <b>Prepared by</b><br>Mohit Tiwari |          | <b>Approved by</b><br>Ravi Jha |      |        |          |
|                                                                                                                            |            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                       | <b>Doc. No.:-</b> PMS SUNPU/PROD/MM/WI-10  |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
|                                                                                                                            |            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                       | <b>Issue Date/Rev. No.</b> 08.07.2025 /00  |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
| <b>Material</b>                                                                                                            | <b>Qty</b> | <b>प्रोसेस</b>                                                                                                      |                                                                                                                                                                                                                                                                                                                       |                                            | <b>कॉम्पोनेन्ट डायग्राम</b>                                                                                                                                                                                                                                 |                                    |          |                                |      |        |          |
| डाइवर बोर्ड में विजुअल डिफेक्ट को देखना<br>Driver board<br>Component :-<br>VISUAL DEFECTS                                  | 1          | <b>S.No.</b>                                                                                                        | <b>निर्देश</b>                                                                                                                                                                                                                                                                                                        |                                            |    |                                    |          |                                |      |        |          |
|                                                                                                                            |            |                                                                                                                     | 1. प्रोसेस में सबसे पहले डाइवर बोर्ड का निरीक्षण करे.<br>1. First of all do the visual inspection of driver board in the visual process.<br><br>2. यदि कोई SMT गायब स्थानांतरित या अतिरिक्त गोंद वाला पाया जाए तो उसे अलग रख दें।<br>If any SMT component found missing,shifting,excess glue on it then put aside it. |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
|                                                                                                                            |            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                       |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
| <b>उपकारी साधन : Equipment Used</b>                                                                                        |            | <b>सावधानी PRECUATION</b>                                                                                           |                                                                                                                                                                                                                                                                                                                       |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
| <b>मैनुअल Manual</b>                                                                                                       |            | कार्य शुरू करने से पहले फिंगर टिप्स और इ.एस.डी बैंड जरूर पहने. Must Wear Finger tips & ESD Band befor starting work |                                                                                                                                                                                                                                                                                                                       |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
| <b>सुरक्षा बिन्दू SAFTY POINTS</b>                                                                                         |            | <b>अनचाहा बर्तन होने पर going any wrong If</b>                                                                      |                                                                                                                                                                                                                                                                                                                       | <b>Revision History</b>                    |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
| ई एस डी बैंड<br>फिंगर टिप्स<br>ESD BAND, FINGER TIP                                                                        |            | लाइन प्रभारी को सूचित करें।                                                                                         |                                                                                                                                                                                                                                                                                                                       | No.                                        | Date                                                                                                                                                                                                                                                        | Reason                             | Approval | No.                            | Date | Reason | Approval |
|                                                                                                                            |            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                       |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |
|                                                                                                                            |            |                                                                                                                     |                                                                                                                                                                                                                                                                                                                       |                                            |                                                                                                                                                                                                                                                             |                                    |          |                                |      |        |          |

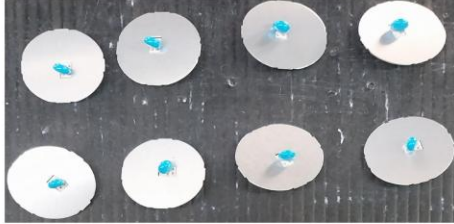
|                                                                                                                              |            |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |
|------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|----------|--------------------------------|------|--------|----------|
|  <b>कार्य निर्देश</b><br>WORK INSTRUCTION |            | <b>MODEL-DOB BULB &amp; ALL MCPCB</b>                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>प्रोसेस :-</b><br>REFLOW VISUAL        |                                                                                                                                                                           | <b>Prepared by</b><br>Mohit Tiwari |          | <b>Approved by</b><br>Ravi Jha |      |        |          |
|                                                                                                                              |            |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Doc. No.:-</b> PMS SUNPU/PROD/WI-09    |                                                                                                                                                                           |                                    |          |                                |      |        |          |
|                                                                                                                              |            |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Issue Date/Rev. No.</b> 08.07.2025 /00 |                                                                                                                                                                           |                                    |          |                                |      |        |          |
| <b>Material</b>                                                                                                              | <b>Qty</b> | <b>प्रोसेस</b>                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           | <b>कॉम्पोनेन्ट डायग्राम</b>                                                                                                                                               |                                    |          |                                |      |        |          |
| MCPCB VISUAL                                                                                                                 | 1          | <b>S.No.</b>                                                                                        | <b>निर्देश</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |   |                                    |          |                                |      |        |          |
|                                                                                                                              |            |                                                                                                     | 1. प्रोसेस में सबसे पहले MCPCB बोर्ड का निरीक्षण करे.<br>1. First of all do the visual inspection of all MCPCB board in the visual process.<br><br>2. अगर कोई SMT कंपोनेन्ट्स मिसिंग या शिफ्टिंग मिले तो उसे अलग कर दे<br>If any SMT component found missing,shifting then put aside it.<br><br>3. MCPCB बोर्ड में निम्न बातों का ध्यान रखें-<br>1 - LED मिसिंग न हो<br>2 - LED शिफ्टिंग नहीं होनी चाइये<br>3 - LED पर डाई नहीं होनी चाइये<br>4 - LED MCPCB पर एक्सेस सोल्डर भी नहीं होना चाइये |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |
|                                                                                                                              |            |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |
|                                                                                                                              |            | <b>सावधानी PRECUATION</b>                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |
| <b>मैनुअल Manual</b>                                                                                                         |            | उत्पादों को उचित स्थानों पर सावधानी से रखें<br>Carefully place the products in Appropriate places., |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |
| <b>सुरक्षा बिन्दू SAFTY POINTS</b>                                                                                           |            | <b>अनचाहा बर्तन होने पर going any wrong If</b>                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>Revision History</b>                   |                                                                                                                                                                           |                                    |          |                                |      |        |          |
| ई एस डी बैंड<br>स्लिपर<br>ESD BAND, SLIPPER                                                                                  |            | अपने लाइन प्रभारी को सूचित करें                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | No.                                       | Date                                                                                                                                                                      | Reason                             | Approval | No.                            | Date | Reason | Approval |
|                                                                                                                              |            |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |
|                                                                                                                              |            |                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                           |                                                                                                                                                                           |                                    |          |                                |      |        |          |



## Assembly:

### 1. Depanelling of MCPCB:

Separating each MCPCB into single pieces for them to be eventually be for further process and to be fixed in the final product.

|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           |                                                                                     |                     |                      |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------|----------------------|
|  <b>कार्य निर्देश</b><br><b>WORK INSTRUCTION</b> |                              | <b>Model :-</b> DOB BULB                                                                                                                                                        | <b>प्रक्रिया Process:-</b><br>एलईडी पीसीबी को डीपनेल करना | Prepared by<br>Sign.                                                                | Checked by<br>Sign. | Approved by<br>Sign. |
|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           | Surajbir                                                                            | Amit pal            | M. Mishra            |
|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           | Doc. No.:                                                                           | PMS SUNPLUPRODW-28  |                      |
|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           | Issue Date/Rev. No.                                                                 | 05.07.2022/00       |                      |
| <b>सामान</b><br><b>Materials</b>                                                                                                  | <b>मात्रा</b><br><b>Qty.</b> | <b>प्रक्रिया Process</b>                                                                                                                                                        |                                                           | <b>गुणवत्ता बिंदु Quality Point</b>                                                 |                     |                      |
| एलईडी पीसीबी को डीपनेल करना                                                                                                       | 1                            | एलईडी पीसीबी को डीपनेल करें जैसा कि चित्र में दिखाया गया है।<br>एलईडी पीसीबी को डीपनेल करते समय ध्यान रखें की कोई लेड डमग न हो।<br>सब असेंबल PCB के बाद अगले चरण में दें        |                                                           | एलईडी पीसीबी को डीपनेल करना और एक एक कर लाइन पर छोड़ना।                             |                     |                      |
|                                                                                                                                   |                              |                                                                                                |                                                           |  |                     |                      |
| <b>उपकरण</b>                                                                                                                      |                              | <b>सावधानी</b>                                                                                                                                                                  |                                                           |                                                                                     |                     |                      |
| मैन्युअल<br>Manual                                                                                                                |                              | फिंगर टिप्स और इ.एस.डी बैंड जरूर पहनें<br>Wear finger tips & ESD band<br>प्रोडक्ट को सावधान पूर्वक उपयुक्त स्थान पर रखे<br>Carefully place the product in the appropriate place |                                                           |                                                                                     |                     |                      |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                                |                              | <b>अंचाहा घांटित होने पर</b><br><b>If any wrong</b>                                                                                                                             |                                                           | <b>Revision History</b>                                                             |                     |                      |
| ESD बैंड, फिंगर टिप<br>ESD BAND, FINGER TIP                                                                                       |                              | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                    |                                                           | No.                                                                                 | Date                | Reason               |
|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           | Approval                                                                            | No.                 | Date                 |
|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           | Reason                                                                              | Approval            | No.                  |
|                                                                                                                                   |                              |                                                                                                                                                                                 |                                                           | Date                                                                                | Reason              | Approval             |

### Process Steps

#### I. Depanelling the LED PCB

Carefully depanel the LED PCBs from the panel as shown in the image.  
 Ensure that each PCB is cleanly separated without causing any damage to the leads or components.

#### II. Handling During Depanelling

While separating the PCBs, pay attention so that no lead or part gets bent or damaged during the process.

#### III. Placement After Depanelling

After depanelling, place each PCB one by one on the production line for the next stage of assembly.

## 2. Insertion of metal oxide varistor on MCPCB:

A **Metal Oxide Varistor (MOV)** is a voltage-dependent resistor used to protect electronic circuits from voltage surges and spikes. In a 9-watt LED bulb, the MOV is typically placed at the AC input stage of the driver circuit, where it acts as a protective component by clamping excessive voltages caused by lightning, switching transients, or other disturbances. Under normal conditions, the MOV remains inactive with high resistance, but when a surge occurs, its resistance drops sharply, allowing it to absorb and divert the excess energy away from sensitive components such as the driver IC and LEDs. This helps in preventing damage, improving reliability, and extending the lifespan of the bulb.

|                                                                                                                                   |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                           |                            |                             |     |      |        |          |
|-----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------|-----|------|--------|----------|
|  <b>कार्य निर्देश</b><br><b>WORK INSTRUCTION</b> | <b>Model :-</b> DOB BULB                                                                                                                                                     | <b>प्रक्रिया Process:-</b><br>एलईडी पीसीबी में MOV फिक्सिंग                                                                                                                                                                                                                                               | <b>Prepared by</b><br>Sign.                                                                                                                               | <b>Checked by</b><br>Sign. | <b>Approved by</b><br>Sign. |     |      |        |          |
|                                                                                                                                   |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                           | Surajbir                                                                                                                                                  | Amit pal                   | M.Mishra                    |     |      |        |          |
|                                                                                                                                   |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                           | Doc. No.:                                                                                                                                                 | PMS SUNPU/PROD/WI-27       |                             |     |      |        |          |
|                                                                                                                                   |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                           | Issue Date/Rev. No.                                                                                                                                       | 05.07.2022/00              |                             |     |      |        |          |
| <b>सामान</b><br><b>Materials</b><br>MOV,<br>एलईडी पीसीबी                                                                          | <b>मात्रा</b><br><b>Qty.</b><br>1<br><br>1                                                                                                                                   | <b>प्रक्रिया Process</b><br>एलईडी पीसीबी में MOV ध्यान से फिक्स करें जैसा कि चित्र में दिखाया गया है।<br>MOV को एमसीपीसीबी में फिक्सिंग के दौरान गैप और टेढ़ा का ध्यान रखें।<br>अब एमसीपीसीबी को अगले चरण में दें।<br> | <b>गुणवत्ता बिंदु Quality Point</b><br>सुनिश्चित करें कि MOV पीसीबी में ठीक से फिक्स होना चाहिए।<br>LED PCB में MOV फिक्सिंग के दौरान गैप ठीक होनी चाहिए। |                            |                             |     |      |        |          |
| <b>उपकरण Equipment</b>                                                                                                            | <b>सावधानी PRECAUTION</b>                                                                                                                                                    |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                           |                            |                             |     |      |        |          |
| <b>मैन्युअल</b><br>Manual                                                                                                         | फिंगर टिप्स और इ.एस.डी बैंड जरूर पहने<br>Wear finger tips & ESD band<br>प्रोडक्ट को सावधानी से उपयुक्त स्थान पर रखें<br>Carefully place the product in the appropriate place |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                           |                            |                             |     |      |        |          |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                                | <b>अंचाहा घाटित होने पर</b><br>If any wrong                                                                                                                                  | <b>Revision History</b>                                                                                                                                                                                                                                                                                   |                                                                                                                                                           |                            |                             |     |      |        |          |
| ESD बैंड, फिंगर टिप<br>ESD BAND, FINGER TIP                                                                                       | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                 | No.                                                                                                                                                                                                                                                                                                       | Date                                                                                                                                                      | Reason                     | Approval                    | No. | Date | Reason | Approval |
|                                                                                                                                   |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                           |                            |                             |     |      |        |          |
|                                                                                                                                   |                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                           |                                                                                                                                                           |                            |                             |     |      |        |          |

## Process Steps

### I. Fixing the MOV on the PCB

Secure the MOV carefully on the LED PCB as shown in the image.

While fixing the MOV on the MCPCB (Metal Core PCB), ensure that:

- The MOV is properly aligned.
- There is no tilt or misplacement.
- Maintain a proper gap around the MOV during placement.



## II. Inspection After Fixing

Confirm that the MOV is firmly fixed to the PCB.

Verify that the gap during fixing is consistent and as per standard requirements before moving the MCPCB to the next production stage.

## 3. Fuse insertion on MCPCB:

A **fuse** in a 9-watt LED bulb serves as a **safety device** designed to protect the internal circuitry from overcurrent or short circuit conditions. It is usually placed at the **AC input line** before the driver circuit. Under normal operation, the fuse allows current to pass through, but if the current exceeds a safe limit due to a fault—such as a shorted component or surge—it blows (opens the circuit), cutting off the power supply and preventing further damage or fire hazards. The fuse ensures the safety of both the bulb and the connected electrical system.

|                                                                                                                        |  |                                                                                                                                                                              |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |
|------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------|--|---------------------------------|------------------------------------------------------------|-----------------------------------|----------|-------------------------------|------|--------|----------|
| <br>कार्य निर्देश<br>WORK INSTRUCTION |  | Model :- DOB BULB                                                                                                                                                            |                                                                                                   | प्रक्रिया Process:-<br>एलईडी पीसीबी में Fuse फिक्सिंग |  | Prepared by<br>Sign. Surajbir   |                                                            | Checked by<br>Sign. Amit pal      |          | Approved by<br>Sign. M.Mishra |      |        |          |
|                                                                                                                        |  |                                                                                                                                                                              |                                                                                                   |                                                       |  | Doc. No.:- PMS SUNPU/PROD/WI-28 |                                                            | Issue Date/Rev. No. 05.07.2022/00 |          |                               |      |        |          |
| सामान<br>Materials<br>Fuse,<br>एलईडी पीसीबी                                                                            |  | मात्रा<br>Qty.                                                                                                                                                               | प्रक्रिया Process                                                                                 |                                                       |  |                                 | गुणवत्ता बिंदु Quality Point                               |                                   |          |                               |      |        |          |
|                                                                                                                        |  | 1                                                                                                                                                                            | एलईडी पीसीबी में Fuse ध्यान से फिक्स करें जैसा कि चित्र में दिखाया गया है।                        |                                                       |  |                                 | सुनिश्चित करें कि Fuse पीसीबी में ठीक से फिक्स होना चाहिए। |                                   |          |                               |      |        |          |
|                                                                                                                        |  | 1                                                                                                                                                                            | Fuse को एमसीपीसीबी में फिक्सिंग के दौरान गैप का ध्यान रखें।<br>अब एमसीपीसीबी को अगले चरण में दें। |                                                       |  |                                 | LED PCB में Fuse फिक्सिंग के दौरान गैप ठीक होनी चाहिए।     |                                   |          |                               |      |        |          |
|                                                                                                                        |  |                                                                                           |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |
| उपकरण Equipment                                                                                                        |  | सावधानी PRECAUTION                                                                                                                                                           |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |
| मैन्युअल<br>Manual                                                                                                     |  | फिंगर टिप्स और इ.एस.डी बैंड जरूर पहने<br>Wear finger tips & ESD band<br>प्रोडक्ट को सावधानी से उपयुक्त स्थान पर रखें<br>Carefully place the product in the appropriate place |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |
| सुरक्षा बिंदु SAFETY POINTS                                                                                            |  | अंचाहा घांटें होने पर<br>If any wrong                                                                                                                                        |                                                                                                   |                                                       |  | Revision History                |                                                            |                                   |          |                               |      |        |          |
| ESD बैंड, फिंगर टिप<br>ESD BAND, FINGER TIP                                                                            |  | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                 |                                                                                                   |                                                       |  | No.                             | Date                                                       | Reason                            | Approval | No.                           | Date | Reason | Approval |
|                                                                                                                        |  |                                                                                                                                                                              |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |
|                                                                                                                        |  |                                                                                                                                                                              |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |
|                                                                                                                        |  |                                                                                                                                                                              |                                                                                                   |                                                       |  |                                 |                                                            |                                   |          |                               |      |        |          |

## Process Steps

### I. Fixing the Fuse on the PCB

Carefully fix the fuse onto the LED PCB as shown in the reference image.

Ensure the fuse is properly inserted and firmly seated in the correct position.

## II. Maintain Proper Gap During Fixing

While fixing the fuse on the MCPCB (Metal Core PCB), maintain the proper spacing and avoid any misalignment or bending of leads.

### III. Move to Next Stage

After the fuse is correctly fixed and inspected, pass the MCPCB on to the next stage of the assembly process.

#### 4. Housing:

The **housing** of an LED bulb serves as the outer protective shell that holds and supports all internal components, such as the light source, electronic driver, and heat management parts. It is typically made from materials that offer both **mechanical strength** and **thermal stability**, allowing it to withstand heat generated during operation while also protecting internal parts from dust, moisture, and physical damage. Additionally, the housing ensures **electrical insulation**, supports **safe handling**, and helps in the proper **dissipation of heat**, contributing to the overall efficiency and longevity of the bulb. The MCPCB is fitted on the housing by automated machines but still to be sure and to reduce any loose fittings they are once again pressed by hand.

|                                                                                                                                   |                                                                                                                                                                                                     |                                                                                      |                                                                                                                                                     |                      |             |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-------------|
|  <b>कार्य निर्देश</b><br><b>WORK INSTRUCTION</b> | <b>Model :- DOB BULB</b><br><b>प्रक्रिया Process:-</b><br><b>हाउसिंग में एलईडी पीसीबी फिक्सिंग</b>                                                                                                  |                                                                                      | Prepared by                                                                                                                                         | Checked by           | Approved by |
|                                                                                                                                   |                                                                                                                                                                                                     |                                                                                      | Sign.                                                                                                                                               | Sign.                | Sign.       |
|                                                                                                                                   |                                                                                                                                                                                                     |                                                                                      | Surajbir                                                                                                                                            | Amit pal             | M.Mishra    |
|                                                                                                                                   |                                                                                                                                                                                                     |                                                                                      | Doc. No.:                                                                                                                                           | PMS SUNPLUPROD/WS-30 |             |
|                                                                                                                                   |                                                                                                                                                                                                     |                                                                                      | Issue Date/Rev. No.                                                                                                                                 | 05.07.2022/00        |             |
| <b>सामान</b><br><b>Materials</b>                                                                                                  | <b>मात्रा</b><br><b>Qty.</b>                                                                                                                                                                        | <b>प्रक्रिया Process</b>                                                             | <b>गुणवत्ता बिंदु Quality Point</b>                                                                                                                 |                      |             |
| एलईडी पीसीबी ,<br>हाउसिंग                                                                                                         | 1<br>सबसे पहले एलईडी पीसीबी को हाउसिंग स्लॉट में दबाएं जैसा कि चित्र में दिखाया गया है।<br><br>1<br>उसके बाद हाउसिंग को कन्वेयर लाइन पर ध्यान पूर्वक छोड़ें।<br>इसके बाद अगले चरण के लिए आगे बढ़ें। |   | एलईडी पीसीबी को ठीक से दबाया जाना चाहिए और आवास में फिक्स करना चाहिए।<br><br>आवास स्लॉट में एलईडी पीसीबी दबाने के दौरान कोई एलईडी नहीं टूटनी चाहिए। |                      |             |
| <b>उपयोग किए हुए उपकरण</b><br><b>Used Equipment</b>                                                                               | <b>सावधानी Precaution</b>                                                                                                                                                                           |                                                                                      |                                                                                                                                                     |                      |             |
| <b>मैन्युअल</b><br><b>Manual</b>                                                                                                  | उत्पाद को उचित स्थान पर सावधानी से रखें।<br>Carefully place the product in appropriate place.                                                                                                       |                                                                                      |                                                                                                                                                     |                      |             |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                                |                                                                                                                                                                                                     | <b>अचानक घाटंत हान पर</b><br><b>If any wrong</b>                                     | <b>Revision History</b>                                                                                                                             |                      |             |
| अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                      |                                                                                                                                                                                                     | No.      Date      Reason      Approval      No.      Date      Reason      Approval |                                                                                                                                                     |                      |             |

## Process Steps

### I. Insert the LED PCB into the Housing Slot

Press the LED PCB carefully into the housing slot as shown in the image.

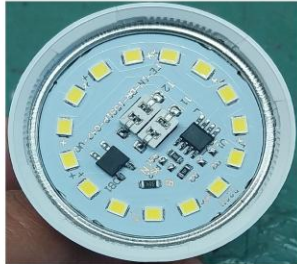
Ensure the PCB is seated properly and fixed firmly in the housing.

## II. Place on Conveyor Line

After fixing, carefully place the housing onto the conveyor line to move to the next process step.

## III. Proceed to Next Stage

Once confirmed that the PCB is fixed correctly without any issues, move forward to the next stage in the assembly process.

| <br>कार्य निर्देश<br>WORK INSTRUCTION |                         | Model :- DOB BULB                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>प्रक्रिया Process :-</b><br>एलईडी पीसीबी को दबाना |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Prepared by<br>Sign.<br>Surajbir                                                                                                                                                                                          | Checked by<br>Sign.<br>Amit pal | Approved by<br>Sign.<br>M.Mishra |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|-----|------|--------|----------|-----|------|--------|----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|                                                                                                                        |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                      | Doc. No.: PMS SUNP/PROD/WI-31<br>Issue Date/Rev. No.: 05.07.2022/00                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                           |                                 |                                  |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>सामान Materials</b><br>एलईडी लैंप                                                                                   | <b>मात्रा Qty.</b><br>1 | <b>प्रक्रिया Process</b><br>सबसे पहले हाउसिंग में लगी एलईडी पीसीबी को फिक्चर में रखना है जैसा कि चित्र में दिखाया गया है।<br>अब फिक्चर के दोनों ओर लगे बटन को दबाना है, जैसा कि चित्र में दिखाया गया है।<br>मशीन में दबाव 0.4 mpa/ 4 psi सुनिश्चित करें<br>ध्यान रहे की कोई भी एलईडी पीसीबी मूडी हुई या बेंड हो।<br>इसके बाद लैंप को अगली स्टेज पर दे। <div style="display: flex; justify-content: space-around;">   </div> |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>गुणवत्ता बिंदु Quality Point</b><br>ध्यान रहे की कोई भी एलईडी पीसीबी मूडी हुई या बेंड हो। <div style="text-align: center;">  </div> |                                 |                                  |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>उपयोग किए हुए उपकरण Used Equipment</b><br>मैन्युअल Manual                                                           |                         | <b>सावधानी Precaution</b><br>उत्पादों को उचित स्थानों पर सावधानी से रखें<br>Carefully place the Products in Appropriate places.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                           |                                 |                                  |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                     |                         | <b>अंचाहा घाटित होने पर If any wrong</b><br>अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                      | <b>Revision History</b> <table border="1" style="width: 100%;"> <thead> <tr> <th>No.</th> <th>Date</th> <th>Reason</th> <th>Approval</th> <th>No.</th> <th>Date</th> <th>Reason</th> <th>Approval</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> </tbody> </table> |                                                                                                                                                                                                                           |                                 |                                  | No. | Date | Reason | Approval | No. | Date | Reason | Approval |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| No.                                                                                                                    | Date                    | Reason                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Approval                                             | No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Date                                                                                                                                                                                                                      | Reason                          | Approval                         |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                        |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                           |                                 |                                  |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                        |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                           |                                 |                                  |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                        |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                           |                                 |                                  |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

## Process Steps

### IV. Place the LED PCB with Housing in Fixture

Position the housing with the attached LED PCB into the fixture carefully, as shown in the image.

### V. Operate the Fixture Buttons

Press the buttons on both sides of the fixture as demonstrated in the image.  
Ensure that the machine pressure is properly set to **0.4 MPa (4 psi)** for secure fixing.

### VI. Check PCB Alignment

Make sure no LED PCB is tilted, bent, or misaligned during this process.

### VII. Proceed to Next Stage

Once fixed correctly, move the assembled lamp to the next stage of the production line.

## 5. Testing:

The product is tested for any faults and if it the fault can be repaired.

| <br>कार्य निर्देश<br>WORK INSTRUCTION |                | Model :- DOB BULB                                                                                                                                                                                                                                                                                                                                                                  | प्रक्रिया Process :-<br>BULB का पूर्व परीक्षण |                  | Prepared by                                                                                                                                     | Checked by         | Approved by       |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
|------------------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|-----|------|-------------|-----------|------|---|-------------|-----------|------|---|-------------|-----------|------|--|--|--|
|                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                    |                                               |                  | Sign.                                                                                                                                           | Sign.              | Sign.             |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
|                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                    |                                               |                  | Surajbir                                                                                                                                        | Amit pal           | M.Mishra          |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
|                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                    |                                               |                  | Doc. No. :-                                                                                                                                     | PMS SUNPUPRODWH-32 |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
|                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                    |                                               |                  | Issue Date/Rev. No.                                                                                                                             | 05.07.2022/00      |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| सामान<br>Materials                                                                                                     | मात्रा<br>Qty. | प्रक्रिया Process                                                                                                                                                                                                                                                                                                                                                                  |                                               |                  | गुणवत्ता बिंदु Quality Point                                                                                                                    |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| सब असेम्बल लैम्प                                                                                                       | 1              | कन्वेयर से लैम्प उठाए और एक -एक कर लैम्प की टेस्टिंग करें।<br>लैम्प का परीक्षण करते समय निम्नलिखित बातों का ध्यान रखें।<br>1. लैम्प नॉट ग्लो / लो वाटेज ना हो<br>2. लैम्प फ्लिकर ना कर रहा हो<br>3. लैम्प का पैरामीटर नीचे दी गई तालिका के अनुसार होना चाहिए।                                                                                                                      |                                               |                  | लैम्प के पैरामीटर दिए गए निर्देशों के अनुसार होने चाहिए।<br> |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
|                                                                                                                        |                | <table border="1"> <thead> <tr> <th>Sr.No.</th> <th>Model</th> <th>Power(In 230V AC)</th> <th>PF</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>5W DOB BULB</td> <td>4.5-5.5 W</td> <td>≥0.9</td> </tr> <tr> <td>1</td> <td>7W DOB BULB</td> <td>6.3-7.7 W</td> <td>≥0.9</td> </tr> <tr> <td>1</td> <td>9W DOB BULB</td> <td>8.1-9.9 W</td> <td>≥0.9</td> </tr> </tbody> </table> |                                               |                  | Sr.No.                                                                                                                                          | Model              | Power(In 230V AC) | PF  | 1    | 5W DOB BULB | 4.5-5.5 W | ≥0.9 | 1 | 7W DOB BULB | 6.3-7.7 W | ≥0.9 | 1 | 9W DOB BULB | 8.1-9.9 W | ≥0.9 |  |  |  |
| Sr.No.                                                                                                                 | Model          | Power(In 230V AC)                                                                                                                                                                                                                                                                                                                                                                  | PF                                            |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| 1                                                                                                                      | 5W DOB BULB    | 4.5-5.5 W                                                                                                                                                                                                                                                                                                                                                                          | ≥0.9                                          |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| 1                                                                                                                      | 7W DOB BULB    | 6.3-7.7 W                                                                                                                                                                                                                                                                                                                                                                          | ≥0.9                                          |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| 1                                                                                                                      | 9W DOB BULB    | 8.1-9.9 W                                                                                                                                                                                                                                                                                                                                                                          | ≥0.9                                          |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| उपयोग किए हुए उपकरण<br>Used Equipment                                                                                  |                | सावधानी Precaution                                                                                                                                                                                                                                                                                                                                                                 |                                               |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| पैरामीटर परीक्षक<br>parameter tester                                                                                   |                | ईएसडी बैंड ज़रूर पहने<br>Must wear ESD band<br>उत्पाद को उचित स्थान पर सावधानी से रखें।<br>Carefully place the product in appropriate place.                                                                                                                                                                                                                                       |                                               |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| सुरक्षा बिंदु SAFETY POINTS                                                                                            |                | अंचाहा घाटित होने पर<br>If any wrong                                                                                                                                                                                                                                                                                                                                               |                                               | Revision History |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
| ईएसडी बैंड<br>ESD BAND                                                                                                 |                | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                                                                                                                                                       |                                               | No.              | Date                                                                                                                                            | Reason             | Approval          | No. | Date | Reason      | Approval  |      |   |             |           |      |   |             |           |      |  |  |  |
|                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                    |                                               |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |
|                                                                                                                        |                |                                                                                                                                                                                                                                                                                                                                                                                    |                                               |                  |                                                                                                                                                 |                    |                   |     |      |             |           |      |   |             |           |      |   |             |           |      |  |  |  |

## Process Steps

### 1. Pick Lamps from Conveyor

Pick each lamp from the conveyor one by one for testing.

### 2. Test Each Lamp

During testing, ensure the following:

- The lamp is glowing properly and does not show **no-glow** or **low wattage** conditions.
- The lamp is **not flickering** during operation.
- The lamp's parameters match the specifications given in the table below.

## Parameter Table

| Sr. No. | Model       | Power (at 230V AC) | Power Factor (PF) |
|---------|-------------|--------------------|-------------------|
| 1       | 5W DOB Bulb | 4.5 – 5.5 W        | $\geq 0.9$        |
| 2       | 7W DOB Bulb | 6.3 – 7.7 W        | $\geq 0.9$        |
| 3       | 9W DOB Bulb | 8.1 – 9.9 W        | $\geq 0.9$        |

## 6. Metal Cap:

- First the metal cap is fitted on the housing by hand.

कार्य निर्देश

WORK INSTRUCTION

Model :- DOB BULB

प्रक्रिया Process :-  
हाउसिंग में मेटल कैप फिटिंग  
Metal Cap fitting in housing

Prepared by

Sign.

Surajbir

Checked by

Sign.

Amit pal

Approved by

Sign.

M.Mishra

Doc. No.:

PMS SUNUPRODWH-33

Issue Date/Rev. No.

05.07.2022/00

सामान  
Materials

बी22 कैप,  
सब असेम्बल लैम्प

मात्रा  
Qty.

1

प्रक्रिया Process

कैप को हाउसिंग में रखें और इनपुट वायर को कैप होल से बाहर निकालें जैसा कि चित्र में दिखाया गया है।  
  
1 दोनों इनपुट तारों को B22 कैप होल से बाहर मोड़ें।  
एक्स्ट्रा वायर को कटर की सहायता से कटे जैसा की चित्र में दिखाया।  
इसके बाद अगले चरण के लिए आगे बढ़ें।



गुणवत्ता बिंदु Quality Point

सुनिश्चित करें कि सभी प्रक्रिया के दौरान इनपुट तार टूटना नहीं चाहिए।

उपयोग किए हुए उपकरण  
Used Equipment

मैन्युअल  
Manual

सावधानी Precaution

उत्पाद को उचित स्थान पर सावधानी से रखें।  
Carefully place the product in appropriate place.

सुरक्षा बिंदु SAFETY POINTS

अवांछित घाटत होने पर  
If any wrong  
अपने लाइन प्रभारी को सूचित करें  
Inform your line incharge

Revision History

| No. | Date | Reason | Approval | No. | Date | Reason | Approval |
|-----|------|--------|----------|-----|------|--------|----------|
|     |      |        |          |     |      |        |          |
|     |      |        |          |     |      |        |          |
|     |      |        |          |     |      |        |          |

## Process Steps

### I. Place the Cap in the Housing

Insert the cap into the housing and carefully pull the input wires out through the cap holes as shown in the image.

### II. Bend the Input Wires

Bend both input wires outward through the B22 cap holes.

### III. Trim Extra Wire

Cut any extra wire using a wire cutter, as demonstrated in the image.



#### IV. Proceed to the Next Stage

After completing the above steps, move the unit to the next stage of assembly.

#### Important Note

Ensure that during the entire process, **no input wire should break** or get damaged.

- Crimping of the metal cap is done on the housing making a hole like ring structure around the opening.

|                                                                                                               |                |                                                                                                                                                                                                                                                      |                                                                   |                     |                                                                                                                                 |                                 |                                  |
|---------------------------------------------------------------------------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|
|  निर्देश<br>WORK INSTRUCTION |                | Model :- DOB BULB                                                                                                                                                                                                                                    | <b>प्रक्रिया Process :-</b><br>B22 कैप पंचिंग<br>B22 Cap Punching |                     | Prepared by<br>Sign.<br>Surajbir                                                                                                | Checked by<br>Sign.<br>Amit pal | Approved by<br>Sign.<br>M.Mishra |
|                                                                                                               |                |                                                                                                                                                                                                                                                      |                                                                   | Doc. No.:           | PMS SUNPUPRODWI-35                                                                                                              |                                 |                                  |
|                                                                                                               |                |                                                                                                                                                                                                                                                      |                                                                   | Issue Date/Rev. No. | 05.07.2022/00                                                                                                                   |                                 |                                  |
| सामान<br>Materials                                                                                            | मात्रा<br>Qty. | प्रक्रिया Process                                                                                                                                                                                                                                    |                                                                   |                     | गुणवत्ता बिंदु Quality Point                                                                                                    |                                 |                                  |
| सब असेम्बल लैम्प, पंचिंग मशीन                                                                                 | 1              | यह कार्य निर्देश दिखाता है कि B22 कैप को कैसे पंच करना है।<br>पंचिंग मशीन में दबाव 0.4 mpa/ 4 psi सुनिश्चित करें<br>लैम्प को पंचिंग मशीन में डालें और उसके कैप को मशीन से पंच करें<br>पंचिंग लैप वैसा ही होना चाहिए जैसा कि चित्र में दिखाया गया है। |                                                                   |                     | सुनिश्चित करें कि छिद्रण ठीक से किया जाना चाहिए और B22cap ढीला नहीं होना चाहिए।<br>टोरसन मशीन से समय-समय पर टोरसन की जांच करें। |                                 |                                  |
| उपयोग किए हुए उपकरण<br>Used Equipment                                                                         |                | सावधानी Precaution                                                                                                                                                                                                                                   |                                                                   |                     |                                                                                                                                 |                                 |                                  |
| मैन्युअल<br>Manually                                                                                          |                | उत्पाद को उचित स्थान पर सावधानी से रखें।<br>Carefully place the product in appropriate place.                                                                                                                                                        |                                                                   |                     |                                                                                                                                 |                                 |                                  |
| सुरक्षा बिंदु SAFETY POINTS                                                                                   |                | अचाहा घाटत हान पर<br>If any wrong                                                                                                                                                                                                                    |                                                                   | Revision History    |                                                                                                                                 |                                 |                                  |
|                                                                                                               |                | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                         |                                                                   | No.                 | Date                                                                                                                            | Reason                          | Approval                         |
|                                                                                                               |                |                                                                                                                                                                                                                                                      |                                                                   |                     |                                                                                                                                 |                                 |                                  |
|                                                                                                               |                |                                                                                                                                                                                                                                                      |                                                                   |                     |                                                                                                                                 |                                 |                                  |
|                                                                                                               |                |                                                                                                                                                                                                                                                      |                                                                   |                     |                                                                                                                                 |                                 |                                  |

#### Process Steps

##### I. Prepare the Lamp for Punching

Place the lamp into the punching machine, ensuring the B22 cap is positioned correctly as shown in the image.

##### II. Set Machine Pressure

Make sure the punching machine is set to a pressure of **0.4 MPa (4 psi)** before starting.

##### III. Perform the Punching

Punch the cap using the machine. The punched lamp should match the reference shown in the image.

### Important Notes

- Ensure the punching is done properly and the **B22 cap is not loose** after the process.
- Regularly check torque using a **torque testing machine** to ensure the cap remains secure over time.

### 7. Eyelet fitting/soldering:

The **eyelet** is a small metallic component located at the bottom center of an LED bulb's base. It acts as one of the **main electrical contact points**, enabling the flow of current from the socket into the bulb. Made from a conductive metal, the eyelet ensures a stable and reliable electrical connection. It is securely attached to the base structure and plays a crucial role in the **safe and efficient operation** of the bulb by facilitating proper electrical contact with the power source.

|                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                     |                                                 |                                                                                                                                                                                                                                                                    |                                       |                                         |
|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------|
| <br>कार्य निर्देश<br>WORK INSTRUCTION |                                                                                                                                              | Model :- DOB BULB                                                                                                                                                                                                                                                                                                                                   | प्रक्रिया Process :-<br>बी22 आईलेट पर सोल्डरिंग | Prepared by<br>Sign. _____<br>Surajbir                                                                                                                                                                                                                             | Checked by<br>Sign. _____<br>Amit pal | Approved by<br>Sign. _____<br>M. Mishra |
| Date Recd. _____<br>Issue Date/Rev. No. _____                                                                          |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                     |                                                 | Date Recd. _____<br>Issue Date/Rev. No. _____                                                                                                                                                                                                                      |                                       |                                         |
| सामान<br>Materials                                                                                                     | मात्रा<br>Qty.                                                                                                                               | प्रक्रिया Process                                                                                                                                                                                                                                                                                                                                   |                                                 | गुणवत्ता बिंदु Quality Point                                                                                                                                                                                                                                       |                                       |                                         |
| बी 22 धारक,<br>सोल्डर तार,<br>सब असेम्बल लैम्प                                                                         | 1                                                                                                                                            | वायर फोल्ड करें और B22 सुराख पर ठीक से लगाएं।<br>सोल्डर वायर को B22 कैप होल्डर की सुराख से मिलाएं<br>सुनिश्चित करें कि बी22 कैप की सुराख पर सोल्डर उचित है<br>सोल्डरिंग ऊँची नीची नहीं होनी चाहिए<br>दोनों सोल्डरिंग आंख की तरह बनी थोप में बराबर होनी चाहिए।<br> |                                                 | <br><br> |                                       |                                         |
| उपयोग किए हुए उपकरण                                                                                                    |                                                                                                                                              | सावधानी                                                                                                                                                                                                                                                                                                                                             |                                                 |                                                                                                                                                                                                                                                                    |                                       |                                         |
| मैन्युअल<br>Manually                                                                                                   | ईएसडी बैंड ज़रूर पहने<br>Must wear ESD band<br>उत्पाद को उचित स्थान पर सावधानी से रखें।<br>Carefully place the product in appropriate place. |                                                                                                                                                                                                                                                                                                                                                     |                                                 |                                                                                                                                                                                                                                                                    |                                       |                                         |
| सुरक्षा बिंदु SAFETY POINTS                                                                                            |                                                                                                                                              | अवाही घाटल हान पर<br>If any wrong                                                                                                                                                                                                                                                                                                                   |                                                 | Revision History                                                                                                                                                                                                                                                   |                                       |                                         |
| ESD बैंड, फिंगर टिप<br>ESD BAND, FINGER TIP                                                                            |                                                                                                                                              | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                                                                                                                        |                                                 | No.                                                                                                                                                                                                                                                                | Date                                  | Reason                                  |
|                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                     |                                                 | Approved                                                                                                                                                                                                                                                           | No.                                   | Date                                    |
|                                                                                                                        |                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                     |                                                 | Reason                                                                                                                                                                                                                                                             | Approved                              |                                         |

### Process Steps

#### I. Fold and Position the Wire

Fold the wire neatly and place it properly on the B22 terminal slot.

#### II. Align Solder Wire with Terminal Hole

Match the solder wire to the hole of the B22 cap holder so that it fits correctly before soldering.

#### III. Ensure Proper Soldering

- Confirm that the solder joint on the B22 cap terminal is adequate and secure.
- The soldering must be **smooth and level**—there should be no excess or uneven solder.

#### IV. Check Solder Shape

Both solder joints should form an **even eye-shaped** pattern for a reliable and professional finish.

**Diffuser:**

The **diffuser** in an LED bulb is a translucent or frosted cover that spreads and softens the light emitted by the LEDs. Its main purpose is to **distribute light evenly**, reduce glare, and create a more comfortable and visually pleasing illumination. Typically made from materials like polycarbonate or acrylic, the diffuser also serves as a **protective barrier**, shielding the internal components from dust, moisture, and accidental contact. By improving light dispersion and safety, the diffuser enhances both the **performance and user experience** of the bulb. The diffuser is fitted on the housing after applying the glue on the inner rim of housing.

|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                         |                                                                                      |                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------|---------------------------|
|  <p><b>कार्य निर्देश</b><br/><b>WORK INSTRUCTION</b></p> |  | <p><b>Model :- DOB BULB</b></p>                                                                                                                           | <p><b>प्रक्रिया Process :-</b><br/>हाउसिंग और फिक्सिंग डिफ्यूज़र में ग्लूइंग<br/>Gluing in Housing and fixing diffuser</p>                                                                                                                        |                         | <p>Prepared by</p> <p>Checked by</p> <p>Approved by</p>                              |                           |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                         | <p>Sign.</p>                                                                         | <p>Sign.</p>              |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                         | <p>Surajbir</p>                                                                      | <p>Amit pal</p>           |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                         | <p>M.Mishra</p>                                                                      |                           |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                         | <p>Doc. No.:</p>                                                                     | <p>PMS SUNPUFRODYN-36</p> |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   |                         | <p>Issue Date/Rev. No.</p>                                                           | <p>05.07.2022/00</p>      |
| <p><b>सामान</b><br/><b>Materials</b></p>                                                                                                  |  | <p><b>मात्रा</b><br/><b>Qty.</b></p>                                                                                                                      | <p><b>प्रक्रिया Process</b></p>                                                                                                                                                                                                                   |                         | <p><b>गुणवत्ता बिंदु Quality Point</b></p>                                           |                           |
| <p>सब असेम्बल लैम्प<br/>डिफ्यूज़र, हाउसिंग<br/>और ग्लूइंग</p>                                                                             |  | <p>1<br/>1</p>                                                                                                                                            | <p>डिफ्यूज़र को हाउसिंग में फिट करने से पहले ये सुनिश्चित कर ले,<br/>की कोई भी LED टूटी हुई न हो,<br/>और न ही किसी भी LED गन्दी न हो।<br/><br/>इसके बाद डिफ्यूज़र को हाउसिंग के लॉक में लगाकर ठीक से फिट करें जैसा कि चित्र में दिखाया गया है</p> |                         | <p>डिफ्यूज़र और हाउसिंग के बीच कोई अंतर नहीं होना चाहिए।</p>                         |                           |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                            |                         |  |                           |
| <p><b>उपयोग किए हुए उपकरण</b><br/><b>Used Equipment</b></p>                                                                               |  | <p><b>सावधानी Precaution</b></p>                                                                                                                          |                                                                                                                                                                                                                                                   |                         |                                                                                      |                           |
| <p>मैन्युअल<br/><b>Manually</b></p>                                                                                                       |  | <p>फिंगर टिप्स ज़रूर पहने<br/>Must wear finger tip<br/>उत्पाद को उचित स्थान पर सावधानी से रखें।<br/>Carefully place the product in appropriate place.</p> |                                                                                                                                                                                                                                                   |                         |                                                                                      |                           |
| <p><b>सुरक्षा बिंदु SAFETY POINTS</b></p>                                                                                                 |  | <p><b>अचवाहा घाटंत हान पर</b><br/><b>If any wrong</b></p>                                                                                                 |                                                                                                                                                                                                                                                   | <p>Revision History</p> |                                                                                      |                           |
| <p>फिंगर टिप<br/><b>FINGER TIP</b></p>                                                                                                    |  | <p>अपने लाइन प्रभारी को सूचित करें<br/>Inform your line incharge</p>                                                                                      |                                                                                                                                                                                                                                                   | <p>No.</p>              | <p>Date</p>                                                                          | <p>Reason</p>             |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   | <p>Approval</p>         | <p>No.</p>                                                                           | <p>Date</p>               |
|                                                                                                                                           |  |                                                                                                                                                           |                                                                                                                                                                                                                                                   | <p>Reason</p>           | <p>Approval</p>                                                                      |                           |

## Process Steps

### I. Pre-check Before Fitting

Before fitting the diffuser onto the housing, carefully inspect the LED PCB:

- Ensure **no LED is broken**.
- Ensure **no LED surface is dirty or contaminated**.

## II. Fit the Diffuser

Place the diffuser onto the housing and lock it securely as shown in the reference image.



### Important Notes

- There should be **no gap** between the diffuser and the housing after fitting.

#### 9. Ageing:

**Ageing** in the context of LED bulbs refers to the **controlled testing process** used to evaluate the **performance, stability, and reliability** of the bulb over time. During ageing, the bulb is operated continuously under normal or slightly elevated voltage and temperature conditions for a specific duration (typically several hours to days). This helps to **identify early failures**, stabilize the internal components (like the driver and LEDs), and ensure consistent performance before the product is packaged and shipped. Ageing is a critical quality control step that helps in delivering **reliable and long-lasting** LED bulbs to the market. The ageing in our case is done for four voltages: 120V, 180V, 220V, 260V.

|                                                                                                                           |  |                                                                               |                                                                                                                                                                                                                                                                                         |                                              |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |
|---------------------------------------------------------------------------------------------------------------------------|--|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|-------------|---------------|-----------------|
|  <b>संदेश</b><br><b>WORK INSTRUCTION</b> |  | <b>Model :- DOB BULB</b>                                                      |                                                                                                                                                                                                                                                                                         | <b>प्रक्रिया Process :-</b><br><b>Ageing</b> |             | <b>Prepared by</b><br><b>Sign.</b><br><b>Surajbir</b>                                                                                   | <b>Checked by</b><br><b>Sign.</b><br><b>Amit pal</b> | <b>Approved by</b><br><b>Sign.</b><br><b>M.Mishra</b> |             |               |                 |
|                                                                                                                           |  |                                                                               |                                                                                                                                                                                                                                                                                         |                                              |             | <b>Doc. No.:</b><br><b>PMS SUNPUSH/PRODW-37</b>                                                                                         |                                                      | <b>Issue Date/Rev. No.</b><br><b>05.07.2022/00</b>    |             |               |                 |
| <b>सामान</b><br><b>Materials</b>                                                                                          |  | <b>मात्रा</b><br><b>Qty.</b>                                                  | <b>प्रक्रिया Process</b>                                                                                                                                                                                                                                                                |                                              |             | <b>गुणवत्ता बिंदु Quality Point</b>                                                                                                     |                                                      |                                                       |             |               |                 |
| <b>एलईडी लैंप को एजिंग</b>                                                                                                |  | <b>1</b>                                                                      | <b>लैंप को एजिंग मशीन में लगाए</b><br><b>एजिंग मशीन को चालू करके वोल्टेज दिए गए वोल्टेज जोन के अनुसार रखें।</b><br><br><b>जोन 01 - 120 ± 10V</b><br><b>जोन 02 - 180 ± 10V</b><br><b>जोन 03 - 220 ± 10V</b><br><b>जोन 04 - 260V ± 10V</b><br><b>स्टैंडर्ड एजिंग का समय 01 घंटे का है</b> |                                              |             | <b>हाथों के दस्ताने और चश्मा जरूर पहने।</b><br><br> |                                                      |                                                       |             |               |                 |
| <b>उपयोग किए हुए उपकरण</b>                                                                                                |  | <b>सावधानी Precaution</b>                                                     |                                                                                                                                                                                                                                                                                         |                                              |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |
| <b>मैन्युअल</b>                                                                                                           |  | <b>चश्मा जरूर पहने।</b><br><b>उत्पादों को उचित स्थानों पर सावधानी से रखें</b> |                                                                                                                                                                                                                                                                                         |                                              |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |
| <b>सुरक्षा बिंदु</b>                                                                                                      |  | <b>अंचाहा घाटेत होने पर</b>                                                   |                                                                                                                                                                                                                                                                                         | <b>Revision History</b>                      |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |
|                                                                                                                           |  | <b>अपने लाइन प्रभारी को सूचित करें</b><br><b>Inform your line incharge</b>    |                                                                                                                                                                                                                                                                                         | <b>No.</b>                                   | <b>Date</b> | <b>Reason</b>                                                                                                                           | <b>Approval</b>                                      | <b>No.</b>                                            | <b>Date</b> | <b>Reason</b> | <b>Approval</b> |
|                                                                                                                           |  |                                                                               |                                                                                                                                                                                                                                                                                         |                                              |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |
|                                                                                                                           |  |                                                                               |                                                                                                                                                                                                                                                                                         |                                              |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |
|                                                                                                                           |  |                                                                               |                                                                                                                                                                                                                                                                                         |                                              |             |                                                                                                                                         |                                                      |                                                       |             |               |                 |

### Process Steps

#### I. Place the Lamp in the Aging Machine

Insert the lamp into the aging machine carefully.

#### II. Set the Voltage According to Zones

Turn on the aging machine and set the voltage as per the specified zone requirements:

- **Zone 01:** 120 V ± 10 V
- **Zone 02:** 180 V ± 10 V
- **Zone 03:** 220 V ± 10 V
- **Zone 04:** 260 V ± 10 V

### III. Run the Standard Aging Cycle

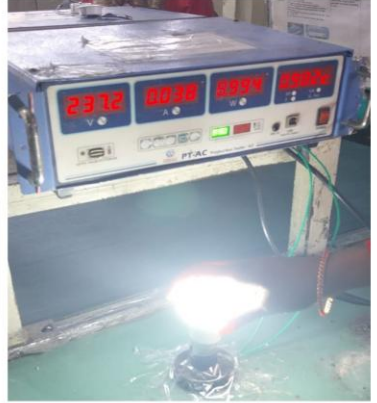
The standard aging time is **1 hour**. Allow the machine to run for the full duration while monitoring the lamps.

#### Precaution

- **Wear gloves and safety glasses at all times** while handling.

### 10. Testing:

The bulbs are again gone for testing before the final quality check.

|  <b>कार्य निर्देश</b><br><b>WORK INSTRUCTION</b> |                       | <b>Model :- DOB BULB</b>                                                                                                                                                                                                                                                                                                                                                           | <b>प्रक्रिया Process :-</b><br><b>BULB का पूर्व परीक्षण</b> |                         | Prepared by         | Checked by                                               | Approved by   |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------|---------------------|----------------------------------------------------------|---------------|-------------------|------|--------|-------------|-----------|------|---|-------------|-----------|------|---|-------------|-----------|------|--------------------------------------------------------------------------------------|--|--|
|                                                                                                                                   |                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                         | Sign.               | Sign.                                                    | Sign.         |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|                                                                                                                                   |                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                         | Surajbir            | Amit pal                                                 | M.Mishra      |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|                                                                                                                                   |                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                         | Doc. No.:           | PMS SUNPU/PROD/WI-38                                     |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|                                                                                                                                   |                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                         | Issue Date/Rev. No. |                                                          | 05.07.2022/00 |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| <b>सामान</b><br>Materials                                                                                                         | <b>मात्रा</b><br>Qty. | <b>प्रक्रिया Process</b>                                                                                                                                                                                                                                                                                                                                                           |                                                             |                         |                     | <b>गुणवत्ता बिंदु Quality Point</b>                      |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| सब असेम्बल लैम्प                                                                                                                  | 1                     | कन्वेयर से लैम्प उठाएं और एक -एक कर लैम्प को टेस्टिंग करें।<br>लैम्प का परीक्षण करते समय निम्नलिखित बातों का ध्यान रखें।<br>1. लैम्प नॉट ग्लो / लो वाटेज ना हो<br>2. लैम्प फ्लिकर ना कर रहा हो<br>3. लैम्प का पैरामीटर नीचे दी गई तालिका के अनुसार होना चाहिए।                                                                                                                     |                                                             |                         |                     | लैम्प के पैरामीटर दिए गए निर्देशों के अनुसार होने चाहिए। |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|                                                                                                                                   |                       | <table border="1"> <thead> <tr> <th>Sr.No.</th> <th>Model</th> <th>Power(In 230V AC)</th> <th>PF</th> </tr> </thead> <tbody> <tr> <td>1</td> <td>5W DOB BULB</td> <td>4.5-5.5 W</td> <td>≥0.9</td> </tr> <tr> <td>1</td> <td>7W DOB BULB</td> <td>6.3-7.7 W</td> <td>≥0.9</td> </tr> <tr> <td>1</td> <td>9W DOB BULB</td> <td>8.1-9.9 W</td> <td>≥0.9</td> </tr> </tbody> </table> |                                                             |                         |                     | Sr.No.                                                   | Model         | Power(In 230V AC) | PF   | 1      | 5W DOB BULB | 4.5-5.5 W | ≥0.9 | 1 | 7W DOB BULB | 6.3-7.7 W | ≥0.9 | 1 | 9W DOB BULB | 8.1-9.9 W | ≥0.9 |  |  |  |
| Sr.No.                                                                                                                            | Model                 | Power(In 230V AC)                                                                                                                                                                                                                                                                                                                                                                  | PF                                                          |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| 1                                                                                                                                 | 5W DOB BULB           | 4.5-5.5 W                                                                                                                                                                                                                                                                                                                                                                          | ≥0.9                                                        |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| 1                                                                                                                                 | 7W DOB BULB           | 6.3-7.7 W                                                                                                                                                                                                                                                                                                                                                                          | ≥0.9                                                        |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| 1                                                                                                                                 | 9W DOB BULB           | 8.1-9.9 W                                                                                                                                                                                                                                                                                                                                                                          | ≥0.9                                                        |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| <b>उपयोग किए हुए उपकरण</b><br>Used Equipment                                                                                      |                       | <b>सावधानी Precaution</b>                                                                                                                                                                                                                                                                                                                                                          |                                                             |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| पैरामीटर परीक्षक<br>parameter tester                                                                                              |                       | ईएसडी बैंड ज़रूर पहनें<br>Must wear ESD band<br>उत्पाद को उचित स्थान पर सावधानी से रखें।<br>Carefully place the product in appropriate place.                                                                                                                                                                                                                                      |                                                             |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                                |                       | <b>अंचाहा घाटित होने पर</b><br>If any wrong                                                                                                                                                                                                                                                                                                                                        |                                                             | <b>Revision History</b> |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
| ईएसडी बैंड<br>ESD BAND                                                                                                            |                       | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                                                                                                                                                       |                                                             | No.                     | Date                | Reason                                                   | Approval      | No.               | Date | Reason | Approval    |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|                                                                                                                                   |                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |
|                                                                                                                                   |                       |                                                                                                                                                                                                                                                                                                                                                                                    |                                                             |                         |                     |                                                          |               |                   |      |        |             |           |      |   |             |           |      |   |             |           |      |                                                                                      |  |  |

#### Process Steps

##### I. Pick Lamps from the Conveyor

Lift the lamps one by one from the conveyor for individual testing.

##### II. Test Each Lamp Carefully

During testing, make sure of the following:

- The lamp should glow properly; it must **not show no-glow or low wattage** conditions.
- The lamp **should not flicker** during operation.
- The lamp's parameters must match the specifications given in the table below.

## Testing Parameters

| Sr. No. | Model       | Power (at 230 V AC) | Power Factor (PF) |
|---------|-------------|---------------------|-------------------|
| 1       | 5W DOB Bulb | 4.5 – 5.5 W         | ≥ 0.9             |
| 2       | 7W DOB Bulb | 6.3 – 7.7 W         | ≥ 0.9             |
| 3       | 9W DOB Bulb | 8.1 – 9.9 W         | ≥ 0.9             |

### 11. Laser printing:

After the ageing process, the brand name “**COLORHOME**” is printed on the bulb using a **laser printing machine**. The machine is programmed with precise specifications, including the **font style, dimensions**, and **exact placement** on the bulb surface to ensure consistent and high-quality branding.

|                                                                                                                             |                              |                                                                                                                                                                                                              |                                                                                                |                                                                                                                                                                                |                                                      |                                                       |          |     |      |        |          |
|-----------------------------------------------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------|----------|-----|------|--------|----------|
|  <b>नेर्देश</b><br><b>WORK INSTRUCTION</b> |                              | <b>Model :- DOB BULB</b>                                                                                                                                                                                     | <b>प्रक्रिया Process :-</b><br><b>लैंप पर लेज़र प्रिंटिंग</b><br><b>Laser Printing on Bulb</b> | <b>Prepared by</b><br><b>Sign.</b><br><b>Surajbir</b>                                                                                                                          | <b>Checked by</b><br><b>Sign.</b><br><b>Amit pal</b> | <b>Approved by</b><br><b>Sign.</b><br><b>M.Mishra</b> |          |     |      |        |          |
|                                                                                                                             |                              |                                                                                                                                                                                                              |                                                                                                | <b>Doc. No.:</b> PMS SURPU/PROD/WI-39<br><b>Issue Date/Rev. No.</b> 05.07.2022/00                                                                                              |                                                      |                                                       |          |     |      |        |          |
| <b>सामान</b><br><b>Materials</b>                                                                                            | <b>मात्रा</b><br><b>Qty.</b> | <b>प्रक्रिया Process</b>                                                                                                                                                                                     |                                                                                                | <b>गुणवत्ता बिंदु Quality Point</b>                                                                                                                                            |                                                      |                                                       |          |     |      |        |          |
| असेम्बल लैंप और प्रिंटिंग मशीन<br>Assembled Lamp & Printing Machine                                                         | 1                            | निर्देश के अनुसार लेजर प्रिंटिंग करें।<br><br>लेजर प्रिंटिंग को आर्टवर्क से मिलाएं फिर प्रिंटिंग करें।<br><br>लेजर प्रिंटिंग मंथ कोड पर होगी।<br><br>आवास पर लेजर प्रिंटिंग जैसा कि चित्र में दिखाया गया है। |                                                                                                | प्रिंटिंग को दिए गए आर्टवर्क से चेक करें।<br><br>सुनिश्चित करें कि छपाई स्पष्ट और पठनीय होनी चाहिए।                                                                            |                                                      |                                                       |          |     |      |        |          |
| <b>उपयोग किए हुए उपकरण</b><br><b>Used Equipment</b>                                                                         |                              | <b>सावधानी Precaution</b>                                                                                                                                                                                    |                                                                                                | <br> |                                                      |                                                       |          |     |      |        |          |
| लेजर प्रिंटिंग मशीन<br>Laser Printing Machine                                                                               |                              | उत्पादों को उचित स्थानों पर सावधानी से रखें<br>Carefully place the Products in Appropriate places.                                                                                                           |                                                                                                |                                                                                                                                                                                |                                                      |                                                       |          |     |      |        |          |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                          |                              | <b>अचानक घाटित होने पर</b><br><b>If any wrong</b>                                                                                                                                                            |                                                                                                | <b>Revision History</b>                                                                                                                                                        |                                                      |                                                       |          |     |      |        |          |
|                                                                                                                             |                              | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                 |                                                                                                | No.                                                                                                                                                                            | Date                                                 | Reason                                                | Approval | No. | Date | Reason | Approval |
|                                                                                                                             |                              |                                                                                                                                                                                                              |                                                                                                |                                                                                                                                                                                |                                                      |                                                       |          |     |      |        |          |

## Process Steps

### I. Perform Laser Printing as per Instructions

Follow the given guidelines and set up the laser printer accordingly.

### II. Match with Artwork

Align the laser printing layout with the provided artwork before starting the printing process.

### III. Print Month Code

The laser printing should be done on the month code area as specified.

### IV. Print on Housing

Carry out the laser printing on the housing exactly as shown in the reference image.

#### Important Notes

- Verify the print with the given artwork to ensure it matches correctly.
- Ensure the printed text is **clear and legible**.

## 12. Visual Inspection and Cleaning:

Each bulb undergoes a visual inspection to check for any marks, dust, or surface contamination. If any imperfections are detected, the bulbs are carefully cleaned to ensure a clean and presentable finish.

|                                                                                                                |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                           |                                               |      |                                                                                                                                                                               |                                 |                                  |      |        |          |
|----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------|------|--------|----------|
| <br>प्रीत<br>WORK INSTRUCTION | Model :- DOB BULB                                                                                  |                                                                                                                                                                                                                                                                                                                                                           | प्रक्रिया Process :-<br>सफाई और उद्देश्य जांच |      | Prepared by<br>Sign.<br>Surajbir                                                                                                                                              | Checked by<br>Sign.<br>Amit pal | Approved by<br>Sign.<br>M.Mishra |      |        |          |
|                                                                                                                |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                           |                                               |      | Doc. No.:                                                                                                                                                                     | PMS SUNPU/PROD/WI-49            |                                  |      |        |          |
|                                                                                                                |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                           |                                               |      | Issue Date/Rev. No.                                                                                                                                                           | 05.07.2022/00                   |                                  |      |        |          |
| सामान<br>Materials                                                                                             | मात्रा<br>Qty.                                                                                     | प्रक्रिया Process                                                                                                                                                                                                                                                                                                                                         |                                               |      | गुणवत्ता बिंदु Quality Point                                                                                                                                                  |                                 |                                  |      |        |          |
| एलईडी लैंप                                                                                                     | 1                                                                                                  | सफाई के लिए साफ कपड़े और आईपीए का इस्तेमाल करें।<br>लैंप पर की गयी लेज़र प्रिंटिंग को अच्छे से चेक करना है।<br>ब्लैक स्पॉट को अच्छे से साफ करें।<br>लैंप पर खरोंच नहीं होनी चाहिए।<br>लैंप की फिटिंग की जाँच करें<br>डिफ्यूज़र के फिटमेंट की जाँच करें।<br>डिफ्यूज़र और हाउसिंग के बीच गैप नहीं होना चाहिए।<br>लैंप के हिलने पर कोई आवाज नहीं होनी चाहिए। |                                               |      | <br> |                                 |                                  |      |        |          |
| उपयोग किए हुए उपकरण<br>Used Equipment                                                                          | सावधानी Precaution                                                                                 |                                                                                                                                                                                                                                                                                                                                                           |                                               |      |                                                                                                                                                                               |                                 |                                  |      |        |          |
| मैन्युअल<br>Manual                                                                                             | उत्पादों को उचित स्थानों पर सावधानी से रखें<br>Carefully place the Products in Appropriate places. |                                                                                                                                                                                                                                                                                                                                                           |                                               |      |                                                                                                                                                                               |                                 |                                  |      |        |          |
| सुरक्षा बिंदु SAFETY POINTS                                                                                    | अंचाहा घाटित होने पर<br>If any wrong                                                               |                                                                                                                                                                                                                                                                                                                                                           | Revision History                              |      |                                                                                                                                                                               |                                 |                                  |      |        |          |
|                                                                                                                | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                       |                                                                                                                                                                                                                                                                                                                                                           | No.                                           | Date | Reason                                                                                                                                                                        | Approval                        | No.                              | Date | Reason | Approval |
|                                                                                                                |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                           |                                               |      |                                                                                                                                                                               |                                 |                                  |      |        |          |
|                                                                                                                |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                           |                                               |      |                                                                                                                                                                               |                                 |                                  |      |        |          |
|                                                                                                                |                                                                                                    |                                                                                                                                                                                                                                                                                                                                                           |                                               |      |                                                                                                                                                                               |                                 |                                  |      |        |          |

### Process Steps

#### I. Cleaning

- Use a clean cloth and IPA (Isopropyl Alcohol) for wiping and cleaning.
- Carefully remove any black spots from the lamp surface.

#### II. Check Laser Printing

- Inspect the laser printing on the lamp thoroughly.
- Ensure the printing is clear, sharp, and without smudges.

### III. Surface Check

- Confirm there are **no scratches** on the lamp body.

### IV. Fitting Inspection

- Verify that the lamp fitting is proper and secure.
- Check the diffuser fitment carefully.
- There should be **no gap** between the diffuser and the housing.
- When the lamp is shaken gently, **no internal noise** should be heard.

### 13. Packaging:

Each bulb is individually packed into its designated box, after which multiple units are grouped and securely placed into larger cartons or **master boxes** for bulk handling and transportation.

|                                                                                                                    |                       |                                                                                                                                                                                                                                                                                                     |                                                      |                                                                                                                                                                                                                                                                                                                                                       |                                      |                                  |
|--------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|----------------------------------|
|  <b>संदेश</b><br>WORK INSTRUCTION |                       | <b>Model :- DOB BULB</b>                                                                                                                                                                                                                                                                            | <b>प्रक्रिया Process :-</b><br>सफाई और उद्देश्य जांच | Prepared by<br>Sign.<br>Surajbir                                                                                                                                                                                                                                                                                                                      | Checked by<br>Sign.<br>Amit pal      | Approved by<br>Sign.<br>M.Mishra |
|                                                                                                                    |                       |                                                                                                                                                                                                                                                                                                     |                                                      | Doc. No.:<br>Issue Date/Rev. No.                                                                                                                                                                                                                                                                                                                      | PMS SUNPU/PROD/W-41<br>05.07.2022/00 |                                  |
| <b>सामान</b><br>Materials                                                                                          | <b>मात्रा</b><br>Qty. | <b>प्रक्रिया Process</b>                                                                                                                                                                                                                                                                            |                                                      | <b>गुणवत्ता बिंदु Quality Point</b>                                                                                                                                                                                                                                                                                                                   |                                      |                                  |
| एलईडी लैंप                                                                                                         | 1                     | सबसे पहले लैंप पर स्लीव लगाना<br>फिर लैंप पर की गयी लेज़र प्रिंटिंग को अच्छे से चेक करना है।<br>मोनो कार्टन पर डोमिनो प्रिंटिंग भी चेक करना है।<br>मास्टर कार्टन को भी चेक करके ही भरना है , कार्टन कहीं से भी कटा फटा हुआ नहीं होना चाहिए।<br>मास्टर कार्टन में लैंप को ज़िग जग पोजीशन में भरना है |                                                      |     |                                      |                                  |
| <b>उपयोग किए हुए उपकरण</b><br>Used Equipment                                                                       |                       | <b>सावधानी Precaution</b>                                                                                                                                                                                                                                                                           |                                                      |                                                                                                                                                                                                                                                                                                                                                       |                                      |                                  |
| मैन्युअल<br>Manual                                                                                                 |                       | उत्पादों को उचित स्थानों पर सावधानी से रखें<br>Carefully place the Products in Appropriate places.                                                                                                                                                                                                  |                                                      |                                                                                                                                                                                                                                                                                                                                                       |                                      |                                  |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                 |                       | <b>अचानक घाटित होने पर</b><br>If any wrong                                                                                                                                                                                                                                                          |                                                      | Revision History                                                                                                                                                                                                                                                                                                                                      |                                      |                                  |
|                                                                                                                    |                       | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                                                                        |                                                      | No.                                                                                                                                                                                                                                                                                                                                                   | Date                                 | Reason                           |
|                                                                                                                    |                       |                                                                                                                                                                                                                                                                                                     |                                                      | Approval                                                                                                                                                                                                                                                                                                                                              | No.                                  | Date                             |
|                                                                                                                    |                       |                                                                                                                                                                                                                                                                                                     |                                                      | Reason                                                                                                                                                                                                                                                                                                                                                | Approval                             |                                  |

### Process Steps

#### I. Apply Sleeve on Lamp

- First, place the protective sleeve onto each lamp carefully.

#### II. Check Laser Printing on Lamp

- Inspect the laser printing on the lamp thoroughly to ensure it is clear and accurate.

III. **Check Domino Printing on Mono Carton**

- Verify that the domino printing on the mono carton is correct and properly aligned.

IV. **Inspect Master Carton Before Filling**

- Ensure the master carton is in good condition.
- The carton must not be torn, cut, or damaged from any side.

V. **Fill Lamps in Master Carton**

- Arrange the lamps in the master carton in a **zig-zag position** for better stability and protection during transport.

**14. Weight Verification Process for LED Lamp Master Cartons:  
Procedure**

I. **Set the Standard Weight**

Begin by configuring the weighing machine with the standard weight specified for the particular LED lamp model. This ensures the machine has a reference point for verification.

II. **Weigh Each Box Individually**

After setting the standard, place each individual box on the weighing machine one by one. Every box must be checked to confirm it meets the required weight specification.

III. **Monitor Machine Indicators**

- **Green light:** The box weight matches the standard and is within acceptable limits.
- **Red or Yellow light:** The weight is outside the tolerance range and needs to be reviewed or corrected before proceeding.

IV. **Print and Apply the Weight Slip**

When the weight is confirmed as correct, the machine automatically prints a weight slip. This slip must be affixed securely to the master carton as proof that the carton has passed the weight verification step.



|  <b>देश</b><br><b>WORK INSTRUCTION</b> |                              | <b>Model :- DOB BULB</b>                                                                                                                                                                                                                                                                                                                                                           |          | <b>प्रक्रिया Process :-</b><br><b>एलईडी लैंप मास्टर का वेट</b>                                                                                                                                                                                                                                                                                                                                                                                                                   |      | <b>Prepared by</b><br>Sign.                                                                                                                                             | <b>Checked by</b><br>Sign. | <b>Approved by</b><br>Sign. |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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|                                                                                                                         |                              |                                                                                                                                                                                                                                                                                                                                                                                    |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      | Surajbir                                                                                                                                                                | Amit pal                   | M.Mishra                    |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                         |                              |                                                                                                                                                                                                                                                                                                                                                                                    |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      | Doc. No.: PMS SUNP/PROD/WI-42<br>Issue Date/Rev. No.: 06.07.2022/00                                                                                                     |                            |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>सामान</b><br><b>Materials</b>                                                                                        | <b>मात्रा</b><br><b>Qty.</b> | <b>प्रक्रिया Process</b>                                                                                                                                                                                                                                                                                                                                                           |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      | <b>गुणवत्ता बिंदु Quality Point</b>                                                                                                                                     |                            |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| एलईडी लैंप<br>मास्टर का वेट                                                                                             | 1                            | सबसे पहले हमें वेट मशीन में किसी भी मॉडल का वेट सेट करना है।<br>फिर एक - एक बॉक्स का वेट करना है ,<br>वेट करते समय निम्न बातों का विशेष ध्यान रखें -<br>1 - अगर हमारा वेट स्टैंडर्ड के अनुसार सही तो हरी लाइट जलेगी<br>2 - अगर वेट सही नहीं है टोवेइट मशीन में लाल या पिली लाइट जलेगी<br>3 - वेट सही होने पर प्रिंटर वेट स्लिप देगा जो की हमें मास्टर कार्टन के ऊपर लगानी होती है। |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |   |                            |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>उपयोग किए हुए उपकरण</b><br><b>Used Equipment</b>                                                                     |                              | <b>सावधानी Precaution</b>                                                                                                                                                                                                                                                                                                                                                          |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |                                                                                      |                            |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>मैन्युअल</b><br><b>Manual</b>                                                                                        |                              | उत्पादों को उचित स्थानों पर सावधानी से रखें<br>Carefully place the Products in Appropriate places.                                                                                                                                                                                                                                                                                 |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |                                                                                                                                                                         |                            |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| <b>सुरक्षा बिंदु SAFETY POINTS</b>                                                                                      |                              | <b>अंचाहा घाटित होने पर</b><br><b>If any wrong</b>                                                                                                                                                                                                                                                                                                                                 |          | <b>Revision History</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |                                                                                                                                                                         |                            |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|                                                                                                                         |                              | अपने लाइन प्रभारी को सूचित करें<br>Inform your line incharge                                                                                                                                                                                                                                                                                                                       |          | <table border="1"> <thead> <tr> <th>No.</th> <th>Date</th> <th>Reason</th> <th>Approval</th> <th>No.</th> <th>Date</th> <th>Reason</th> <th>Approval</th> </tr> </thead> <tbody> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> <tr><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td><td> </td></tr> </tbody> </table> |      |                                                                                                                                                                         |                            |                             | No. | Date | Reason | Approval | No. | Date | Reason | Approval |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| No.                                                                                                                     | Date                         | Reason                                                                                                                                                                                                                                                                                                                                                                             | Approval | No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Date | Reason                                                                                                                                                                  | Approval                   |                             |     |      |        |          |     |      |        |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
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### 15. Final quality check:

After completion of all manufacturing, assembly, and packaging steps, a final quality inspection is carried out for the entire batch. This ensures that all bulbs meet the required standards, and only then is the product considered fully complete and ready for dispatch.

### 16. Dispatching:

Once the final quality check is completed and approved, the finished products are prepared for **dispatch**. The master cartons are labeled, documented, and loaded for delivery to distribution centers, retailers, or customers, marking the final stage of the production process.



# Comprehensive Technical Overview of Key Tests for 9-Watt LED Bulb Manufacturing

Ensuring that a 9-watt LED bulb complies with industry standards and safety regulations involves a series of rigorous, closely monitored tests. Each test targets a specific aspect of performance, durability, or safety. Collectively, these evaluations guarantee that the finished product will operate reliably in real-world conditions—whether in residential lighting, commercial installations, or areas with unstable power supplies. Understanding these tests provides engineers and designers with valuable insight into how each step contributes to product quality and longevity.

## Lumen Test

The lumen test is one of the most essential evaluations because it directly measures the total light output and efficiency of the 9-watt LED bulb. Using an integrating sphere or comparable calibrated equipment, the bulb is operated continuously for about 45 minutes while sensors record luminous flux and color quality under steady-state conditions. The measured values for this bulb are summarized below:

| Parameter                   | Value              |
|-----------------------------|--------------------|
| Light Output                | 945 lm $\pm$ 10%   |
| Color Temperature           | 6500 K $\pm$ 510 K |
| Initial Efficiency          | 109.67 lm/W        |
| Market Reference Efficiency | 105 lm/W           |

These results confirm that the bulb surpasses typical market standards for luminous efficacy. If readings fall outside these ranges, it may indicate problems such as substandard LED chips, driver miscalibration, or optical misalignment. Engineers rely on this data to refine designs and ensure optimal energy efficiency and visual performance.

## Surge Test

The surge test evaluates how well the 9-watt bulb withstands high-voltage transients caused by lightning, grid switching, or local electrical disturbances. Over the course of one hour, the bulb is subjected to controlled voltage surges from 0.5 kV to 4 kV in 0.5 kV increments. This broad range tests the limits of protection circuitry. Passing the surge test proves that components such as metal oxide varistors (MOVs) and transient suppression circuits can absorb or divert sudden voltage spikes, ensuring safe, stable operation even in regions with inconsistent power supply.

## Driver Testing

The electronic driver is central to an LED bulb's performance, regulating current and protecting the LEDs from fluctuations. Driver testing examines current stability under varying input voltages, monitors heat generation and ripple current, and evaluates repeated surge handling. Protective features—including overvoltage shutdown, thermal cutoffs, and short-circuit safeguards—are also verified. For a 9-watt design, these results demonstrate whether the driver delivers both high efficiency and long operational life.

## Glow Wire Testing

Glow wire testing addresses fire safety, a critical consideration for devices connected to mains electricity. In this test, a pre-heated wire at approximately 650 °C is applied to key areas of the bulb's housing and insulation for 30 seconds. The material must not ignite or must self-extinguish quickly. A successful outcome shows that, even under fault conditions such as overheating, the bulb's enclosure will not contribute to fire propagation. This assurance is especially important for compact 9-watt bulbs used in enclosed fixtures.

## High Pot Test (Dielectric Strength Test)

The high pot test verifies electrical insulation between live parts and surfaces that could be touched by a user. A voltage of 3.75 kV—much higher than normal operating voltage—is applied for a specified period while monitoring for leakage or breakdown. A passing test confirms that the bulb's insulation system will protect users throughout its service life, even under abnormal power conditions, and ensures compliance with international safety standards.

## Switching Test

The switching test determines how well the 9-watt bulb endures frequent on/off cycles, simulating real usage in locations such as corridors or living rooms. The bulb is cycled thousands of times while engineers watch for light degradation, driver instability, or component failure. Industry standards often require testing for 50% of the rated lifetime; for example, a bulb rated for 25,000 hours undergoes about 12,500 switching cycles. Meeting this benchmark ensures that the bulb maintains its performance and reliability over long-term use.

By combining the results of these tests, manufacturers can be confident that every 9-watt LED bulb leaving the production line not only meets its advertised specifications but also delivers safe, efficient, and durable performance throughout its lifetime.

# Learning and Skills Gained During My Internship

## Technical Knowledge and Process Understanding

This internship experience directly connected to my academic studies and future career goals, reinforcing concepts from coursework with hands-on exposure. I gained a detailed understanding of the complete manufacturing workflow for a 9-watt LED bulb, learning how each stage builds on the previous one to deliver a reliable product. From applying solder paste in the Surface Mount Technology (SMT) process to precision component placement and reflow soldering, I observed how automated machinery and manual inspection work together. I also learned to read and interpret the Bill of Materials (BOM) and understand how each component—LED chips, driver ICs, resistors—contributes to the design, electrical performance, and long-term reliability. This comprehensive exposure helped me appreciate the careful planning and coordination needed to ensure consistent quality.

Equally valuable was my introduction to the testing procedures used to confirm product performance and safety. I observed lumen testing to verify brightness and efficiency following **LM-79** guidelines, surge testing for voltage stability in line with **IEC 61000-4-5**, driver testing to monitor current regulation, glow wire testing for fire safety as specified in **IEC 60695-2**, high-pot testing for insulation strength in accordance with **IEC 61010**, and switching tests to evaluate durability over time. Seeing these tests carried out in real settings deepened my understanding of engineering standards and their role in ensuring that every product meets rigorous industry requirements.

## Practical Skills and Industry Exposure

Beyond technical knowledge, I gained valuable practical skills. I learned to operate and monitor machines like glue dispensers and pick-and-place mounting equipment, as well as to identify and correct issues such as solder bridging, misaligned components, and uneven paste application. These experiences strengthened my troubleshooting abilities, enabling me to trace defects back to their sources and apply effective solutions.

I also developed stronger analytical thinking by studying data from testing and production runs, learning to link measurements with potential design or assembly problems. This approach taught me to view each issue not only as an isolated defect but as an opportunity for process improvement. Additionally, preparing detailed notes and process flow diagrams improved my documentation skills and reinforced the importance of precision and consistency in manufacturing practices.

## Interpersonal and Professional Development

Moving from technical skills to interpersonal growth, I gained valuable soft skills that are essential for any engineer. Working closely with experienced engineers and production

teams improved my communication and collaboration skills. I learned how to ask informed questions to clarify complex topics and how to manage my time effectively in a fast-paced manufacturing environment. These experiences increased my confidence in handling real-world engineering challenges and strengthened my ability to work within multidisciplinary teams.

## Conclusion

Through this internship, I built a strong foundation of technical knowledge, hands-on skills, and professional competencies that connect directly to my studies and future engineering career. My time in the design and development department gave me a clear view of how process planning, quality control, and rigorous testing come together to create a high-quality 9-watt LED bulb. I learned to see beyond individual tasks and understand the broader system that ensures reliable, safe, and efficient products.

Overall, this internship offered an invaluable opportunity to observe and participate in real industrial practices. It allowed me to experience how design, engineering, and production integrate into a successful product workflow. The insights and skills I gained will guide me as I continue to grow as an engineer and apply these lessons in both academic projects and future professional roles.

## References

1. ANSI/IES LM-79-19: Approved Method for Electrical and Photometric Measurements of Solid-State Lighting Products.
2. IEC 61000-4-5: Electromagnetic compatibility (EMC) – Testing and measurement techniques – Surge immunity test.
3. IEC 60695-2 Series: Fire hazard testing – Glow-wire test methods.
4. IEC 61010: Safety requirements for electrical equipment for measurement, control, and laboratory use.
5. General LED performance guidelines and BOM analysis from internal documentation and industry practices.